Public Choice
https://doi.org/10.1007/s11127-023-01082-9

Wealth inequality and democracy

Sutirtha Bagchi[1] · Matthew J. Fagerstrom[2]

Received: 28 September 2022 / Accepted: 12 June 2023

Abstract
Scholars have studied the relationship between land inequality, income inequality, and democracy extensively, but have reached contradictory conclusions that have resulted from competing theories and methodologies. However, despite its importance, the effects of wealth inequality on democracy have not been examined empirically. We use a panel dataset of billionaire wealth from 1987 to 2012 to determine the impact of wealth inequality on the level of democracy. We measure democracy using Polity scores, Varieties of Democracy (V-Dem) indices, and the continuous Machine Learning index. We find limited empirical support for the hypothesis that overall wealth inequality or inherited wealth inequality has an impact on democracy. However, we find evidence that politically connected wealth inequality lowers V-Dem and Machine Learning democracy scores. Following Boix (Democracy and redistribution, Cambridge University Press, New York, 2003), we investigate the hypothesis that capital mobility moderates the relationship between wealth inequality and democracy and find evidence that increased capital mobility mitigates the negative impact of politically connected wealth inequality on democracy.

Keywords Wealth inequality · Democratization · Capital mobility

JEL Classification D31 · H11 · P48

1 Introduction

The relationship between inequality and democracy has become increasingly important in the minds of voters, politicians, and academics. Despite this importance, the literature on inequality and democracy has not reached a consensus. Political scientists and economists have brought their expertise to bear on this question with mathematical models, large-N regression analyses, and case studies. However, notwithstanding this extensive body of research, the inequality-democracy nexus is home to conflicting findings in the literature, with Boix (2003) finding evidence of a negative relationship between land inequality and democratization, Ansell and Samuels (2010) reporting a *positive* relationship between

✉ Sutirtha Bagchi
sutirtha.bagchi@villanova.edu

1 Villanova University, Villanova, PA, USA

2 The Wharton School, University of Pennsylvania, Philadelphia, PA, USA

income inequality and democratization, but a *negative* relationship between land inequality and democratic transitions, and Houle (2009) finding that inequality has no relationship with democratic transitions but increases the risk of authoritarian backsliding.

The lack of consensus in the inequality—democratization literature can be attributed to a number of causes. First, the measure of inequality used in the literature varies from paper to paper, with capital's share of income (Houle, 2009, 2016), land-holding inequality (Ansell & Samuels, 2010; Midlarsky, 1992; Ziblatt, 2008), and the Gini index of income inequality (Krieger & Meierrieks, 2016; Krieckhaus et al., 2014), all being used.[1] Each measure has its own advantages. Capital's share of income is useful primarily because it captures inter-group inequality, and therefore measures the distribution of resources between capital and labor (Houle, 2009). Land inequality measures the concentration of the holding of a fixed asset, and can usefully measure one type of wealth inequality. The Gini index is commonly used and more widely available than either capital's share of income or inequality of land holdings. The use of capital's share of income or the Gini index for income may however not be appropriate for answering questions related to wealth inequality. While the use of a measure of inequality based on land holdings as a proxy for wealth inequality is an improvement over these measures, a measure of land inequality is appropriate for low-income, agrarian economies as opposed to wealthier, industrialized countries.

We innovate on the existing literature by using billionaire wealth as a percentage of GDP as our measure of wealth inequality—a measure first introduced in Bagchi and Svejnar (2015) and subsequently expanded in Bagchi et al. (2019). This measure has three main advantages. First, billionaire wealth is a more comprehensive gauge of wealth than holdings of land, meaning that we capture more asset classes. Because land inequality only focuses on one asset, it is less relevant as economies diversify (Kaufman, 2009). An additional advantage of using wealth inequality instead of land inequality is that it allows us to test whether or not capital mobility mitigates the impact of wealth inequality on democracy (Boix, 2003). Because land is vulnerable to expropriation, it does not allow us to test if changes in wealth vulnerability attenuate the relationship between wealth inequality and democracy. Second, our measure represents a more extreme version of inter-group inequality by looking at what the very top of the wealth distribution owns. To the extent that models of democratization emphasize the role of elites in democratization (or the lack thereof), our measure of wealth inequality is likely to better capture the influence of those elites than a measure like the Gini coefficient that captures inequality over the entire distribution. Third, because we undertake the exercise of classifying each billionaire in our sample as self-made or inherited, and separately as politically connected or unconnected, we can disaggregate wealth inequality and comment on the effects of various components of wealth inequality on democracy.[2] By contrast, aggregate measures of inequality, such as the Gini index, do not permit the classification of economic elites, and as the prior literature has shown (Bagchi & Svejnar, 2015; Easterly, 2007), differences in the source of economic inequality can matter significantly for the outcomes we care about.

[1] Krieger and Meierrieks (2016) investigate the relationship between inequality and economic freedom, rather than democracy.

[2] We have four groups of billionaires: self-made and politically unconnected (e.g. Bill Gates), self-made and politically connected (e.g. Russian oligarch, Roman Abramovich), inherited and politically unconnected (e.g. David Rockefeller), and inherited and politically connected (e.g. the sons of former Lebanese Prime Minister Rafik Hariri).

The ability to look at the heterogeneity within economic elites is important because the expected impact of wealth inequality on democracy is *a priori* ambiguous according to the political agency theories. In the simplest renditions of the Meltzer and Richard (1981) model, economic elites in general may operate under the belief that voters will vote to redistribute their wealth and therefore would oppose democracy in order to avoid having redistributive taxes imposed on their wealth (Acemoglu & Robinson, 2006; Krieckhaus et al., 2014).[3] However, as some recent work (Ansell and Samuels (2010) and Albertus and Menaldo (2018)) suggests, the elites are not a monolithic group and may not share identical views on the desirability of democratization. For instance, politically connected billionaires who owe their fortunes to the current regime may prefer the continuation of that regime as any replacement of the existing political regime (or incumbent politicians) may lead to the elimination of their advantages in the economic sphere such as monopoly rights. In contrast, because authoritarian governments can also more easily expropriate the wealth of individuals who are politically unconnected (Mizuno et al., 2017), those unconnected elites may prefer democracy if they view the protection of property rights and rule of law as more credible under a democracy than an autocracy. By classifying billionaires as either politically connected or unconnected, we can comment on the effect of the two types of inequality on measures of democracy.

A second possible reason for the lack of a consensus in this literature is that few papers closely examine the underlying theoretical relationships that supposedly drive the inequality—democratization relationship. Although numerous studies have looked at political agency theories of inequality and democracy, building on the canonical Meltzer and Richard (1981) model of the median voter, these empirical examinations rarely consider other attenuating factors that are likely to moderate that relationship. In contrast, our work examines an important channel—the role of capital mobility as outlined in Boix (2003)—and considers how variation in the extent of capital mobility can moderate the relationship between inequality and democracy.

Given the plethora of papers that theorize about the existence of a relationship between inequality and democracy, we do not seek to offer yet another theoretical model and instead view our contributions as primarily empirical. We construct a panel dataset of 149 countries for the period from 1987–2017 and examine the impact of our measure of wealth inequality using measures of democracy from Polity, the Varieties of Democracy (V-Dem) project, and the continuous Machine Learning index of democracy (Gründler & Krieger, 2021). Extending our analysis through 2017 allows us to examine the beginning of the rise of populist regimes across several existing democracies and the beginning of a "democratic recession" (Diamond, 2015), or "third wave of autocratization" (Lührmann & Lindberg, 2019). Further, we are able to comment on the rise of "democracy without rights" or the decoupling of democracy from liberal norms of politics (Mounk, 2018).[4] Because it takes time for large structural changes to occur, wealth inequality enters into the regression with a five-year lag. We first record democracy scores in 1992, as *Forbes* compiled a global list of billionaires for the first time in 1987.

[3] We could also consider the political agency of poor voters, who may prefer democracy if inequality is high if they think they can influence the choice of policies that will address inequality. Absent such a belief, they may not seek democracy, believing that it is ineffective in resolving inequality (Krieckhaus et al., 2014). Empirically, there is evidence that high levels of inequality depress voter turnout (Dash et al., 2023).

[4] The use of continuous measures of democracy is important for this point; to the extent that fundamental rights to political participation are still present, we might see a decline in the quality of democracy without seeing a transition to outright autocracy.

To preview our results, we find no relationship between inherited wealth inequality and democracy and weak evidence of a negative relationship between overall wealth inequality and democracy. However, we find robust evidence of a negative relationship between politically connected wealth inequality and levels of democracy across the globe and in a sub-sample of countries that are already democratic. Our results are congruent with the literature on the effects of inequality in democratic consolidations, but with the added caveat of showing that the result is caused by billionaires who profited from political connections. Our baseline results suggest that a one standard deviation increase in politically connected wealth inequality (2.3 percent of GDP) lowers V-Dem Electoral Democracy scores by 0.009 points, *ceteris paribus*.[5] We also find strong evidence in favor of the theory of Boix (2003), which argues that the vulnerability of assets to confiscation is what makes inequality harmful for democracy. The negative relationship between politically connected wealth inequality and democracy is robust to most alternative specifications, but is weakened by the exclusion of countries without billionaires.

Our paper proceeds in seven sections. Section 2 situates this paper in the public choice and political economy literature and outlines the theoretical mechanisms through which politically connected wealth inequality may impact democracy. Section 3 describes the data, including our choice of democracy measures and the construction of our wealth inequality measures. Section 4 lays out the empirical approach, Sect. 5 presents the baseline results, and Sect. 6 discusses the results from our robustness tests. Finally, Sect. 7 concludes.

2 Literature review and theoretical mechanisms

In much of the prior literature, inequality can be seen as either a structural determinant of democracy or as a factor impacting the decision making of political agents. In the structural factor theory, income inequality can lead to democratization because inequality is a byproduct of development, while landholding inequality can harm democratization by thwarting modernization. The political agency theory is largely based around the Median Voter Theorem as expressed in Meltzer and Richard (1981). Building on this theory, an implication that is drawn out both in Acemoglu and Robinson (2006) and Boix (2003) is that wealthy elites block democratization because they are fearful of redistribution. We build upon this line of work through our use of the *Forbes* billionaire lists to construct a novel measure of wealth inequality and examine the effect of inequality on various measures of democracy. More importantly though, unlike traditional survey-based measures of inequality, because we construct our measure of inequality using micro-data on the individuals who are at the very top of the wealth distribution for a given country, we can classify these individuals as either inherited or self-made and, separately as, either politically connected or unconnected. This data construction then allows us to empirically test the importance of different classes of economic elites, in contrast to most papers in the literature that do not—and cannot—make such distinctions. We proceed with this split because our view is that elites are not one monolithic group, and even separating them into economic and political elites misses the rich heterogeneity within economic elites and differences in the ways in which they acquire and/ or preserve their wealth.

[5] For this result, we are using politically connected billionaire wealth as a share of GDP.

Recognizing the importance of heterogeneity in terms of examining the antecedents of democracy is a well-established tradition within Public Choice. For example, in a paper examining the relationship between Islam and democracy, Rowley and Smith (2009) argue that lumping all Muslim-majority countries together is inappropriate. Instead they suggest that scholars consider the unique history and political institutions of each Muslim-majority country to form informed opinions about the degree to which democracy is likely to take root (or not) in each such country. Likewise, taking the idea spelled out in Wintrobe (1990) seriously that the impact of economic growth on repression depends on the type of autocracy considered, Islam and Winer (2004) provides empirical evidence in support of that thesis thereby distinguishing between different kinds of autocracies that had been hitherto lumped together. We follow this broad tradition within Public Choice by discriminating between various classes of economic elites and re-examining the relationship between inequality and democracy in hopes of shedding new light on an old question.

Our paper also directly builds on some recent work that emphasizes the importance of a newly emergent class of economic elites to the establishment of a democracy. In this genre of work, Ansell and Samuels (2010) are among the first to argue that income inequality can lead to democratization because as the economy develops, a new economic elite arises that demands property rights protections and participation in government. In their view rising economic groups such as an urban financial bourgeoisie, finding themselves politically disenfranchised, struggle to obtain credible commitments against expropriation of their income and assets by the autocratic governing elite.[6] They eventually come to a contractarian bargain with the political elite that slowly but steadily leads to democratization.

Speaking more directly to our work that emphasizes the heterogeneity among the economic elites, Albertus and Menaldo (2018) argue that transitions to democracy tend to come about when a new economic elite arises that is unconnected to the prior regime. While the older economic elite may be connected to the autocrat, with their fortunes tied up with those of the authoritarian regime,[7] the same does not hold true for the new economic elite. This group of individuals are more likely to owe their fortunes to success in the economic marketplace and are arguably wary of having their wealth expropriated by the autocrat. Accordingly, this group comprising the new economic elites are likely to support attempts at democratization, and the political incumbents, seeing the writing on the wall, might choose to bargain with them in the hopes of managing the transition to a democracy that still retains some of their traditional advantages (Albertus & Menaldo, 2018).

The implication of this body of work follows naturally for us and our question of interest; it suggests that countries with high levels of politically connected wealth inequality are less likely to democratize or are more likely to stay autocratic because there is less pressure from an outsider economic elite. In contrast, as an outsider economic elite group emerges, perhaps because of changes in the structure of the economy or because of an exogenous economic shock that forces the country to open itself up, the dynamics start changing. As the wealth of this outsider group grows, they can begin to exert more influence. Left unchecked, the possibility exists that this new emergent group could supplant

[6] For a similar exposition of how a rising bourgeoisie class led to a push towards dismantling rent-seeking and demands for political participation in the case of *Ancien Régime* France, see Ekelund and Thornton (2020).

[7] Tullock (1986) anticipates this point, noting that in the typical dictatorship or monarchy, it is common to find a great deal of rent seeking activity and the "granting of monopolies of one sort or another to friends of the ruler is very common and one of the major forms of enterprise is to 'court' the ruler in hopes of getting such special privileges."

the political incumbents. Rather than allow that process to unfold, the incumbent political elite might find it rational to come to the bargaining table, agreeing to a managed transition to democracy that locks in some of their pre-existing advantages. Such processes are less likely to play out in societies where nearly all the economic elites owe their success to connections with the political incumbent, as was the case in Suharto's Indonesia during the 1980s and 1990s until his resignation in 1998, or Russia under Boris Yeltsin in the 1990s and subsequently under Vladimir Putin since 2000.

Put differently, our claim is that simply knowing that a country has a high level of wealth inequality is not enough to draw inferences about its path toward democratization; we need to have some sense of whether that inequality can be traced to elites who exploit their political connections and are close to the regime or if the inequality comes about because of the natural play of market forces. In doing so, we follow Easterly (2007) who notes that: "One confusion in the theoretical and empirical analysis of inequality is between what we could call structural inequality and market inequality. Structural inequality reflects such historical events as conquest, colonization, slavery, and land distribution by the state or colonial power; it creates an elite by means of these non-market mechanisms. Market forces also lead to inequality, but just because success in free markets is always very uneven across different individuals, cities, regions, firms, and industries....Only structural inequality is unambiguously bad for subsequent development in theory; market inequality has ambiguous effects—it could have some of the adverse effects cited in the above models, but eliminating it would obviously have negative incentive effects." We could not agree more. Accordingly, we undertake a similar exercise as in Easterly (2007); by considering the heterogeneity *between* economic elites, we are able to offer some new insights. In contrast, most papers examining the relationship between inequality and democracy fail to distinguish between structural and market inequality and their failure to consider what gives rise to inequality in the first place perhaps explains the morass of findings in this literature alluded to in the Introduction.

By grounding our measure of wealth inequality at the level of the individual billionaire, we also take the principle of methodological individualism seriously and are better able to understand what types of inequality might negatively impact democracy and what types are more benign. To provide a concrete example, it is unclear why we would expect wealth inequality stemming from the fortunes of say, Ms. Abigail Johnson, the current CEO and Chairperson of Fidelity Investments, to be similar to the effects resulting from those of say, Mr. Prajogo Pangestu, the Indonesian billionaire owner of Barito Pacific. Ms. Johnson, who inherited her stake in the investment giant, Fidelity Investments, from her father, Edward "Ned" Johnson III, has chosen to entirely stay away from the public eye and her wealth can be traced to her family's successful stewardship of Fidelity Investments over the years. In contrast, Mr. Pangestu's rise is seen as inextricably linked with the rise of President Suharto, who served as Indonesia's President for approximately 32 years between 1967 and 1998.[8] As is well-documented, Mr. Pangestu's proximity to President Suharto was instrumental to the success of his firm, Barito Pacific. For instance, a 1993 article in the Wall Street Journal notes that: "Mr. Prajogo, Mr. Bambang and their partners also stand to gain from a change of government policy last year that allowed them to proceed with their postponed 1.6 billion PT Chandra Asri petrochemical project, the products of

[8] See also: https://www.nytimes.com/1993/10/01/business/worldbusiness/IHT-a-giant-joins-jakarta-exchange.html which describes the Indonesian business environment in the following dire terms: "All the really big conglomerates in Indonesia have strong political connections. That is how they get favorable contracts and concessions from the government and loans from state banks."

which will be protected by steep new tariffs on imports." Likewise, the same article notes that the "Barito Group of companies, for example, is the state banking system's largest borrower, with loans of more than 1 billion outstanding, and benefits from an unusually attractive 1992 debt rescheduling that stretched repayment periods on about 460 million in timber industry borrowings into the next century." Individuals like Mr. Pangestu and others such as Mr. Liem Sioe Liong of the Salim Group, who were engaging in rent seeking and obtaining favorable terms from state-owned banks or regulators during the Suharto regime, would have naturally faced stronger incentives to support the regime than individuals who were not reliant upon similar rent-seeking to grow their fortunes. It would be strange for us to assume that individuals as diverse as Ms. Johnson and Mr. Pangestu would share similar preferences for institutional arrangements in their respective countries just because they happen to find a spot at the very top of the wealth distribution in those countries.

Thus rather than club all elites, even all economic elites, as one class, we dig deeper by considering the paths followed by these different individuals in their rise to riches. Doing so lets us distinguish between the different institutional arrangements favored by each individual billionaire based on the incentives that they face as individuals in either continuing with the existing regime or in advocating for change. In that regard we follow Buchanan and Tullock (1999) by thinking about how the costs faced by individuals in the political sphere shape their preferences for institutions and the extent of collective action.

Moving away from individuals to countries, comparing the United States, Indonesia, and Russia also proves to be useful in establishing our claim that differences in the source of inequality matter. Although billionaire wealth is high in the United States, rising to 10 percent of GDP by 2012, V-Dem electoral democracy scores fluctuate between 0.83 and 0.90 over the period from 1987 to 2017. In contrast to this high level of overall inequality, politically connected wealth inequality in the U.S. never rises above 0.15 percent of GDP. The lack of politically connected wealth, as classified by us, perhaps explains why despite relatively high levels of wealth inequality compared to say, continental Europe, scores on democracy indices for the United States are comparable to (or higher than) those in continental Europe. For instance in 2012, wealth inequality in the three largest European countries—France, Germany, and the United Kingdom—averaged about 5.5 percent of GDP, compared to the U.S. figure of 10.0 percent. Despite that fact, the United States scored the highest possible 10 points on the Polity index, alongside Germany and the United Kingdom, and 1 point *higher than* France. Although hardly dispositive, these observations are consistent with the argument that it is politically connected wealth inequality that matters more for democracy, rather than overall wealth inequality.

In contrast to the United States, all of the billionaires who show up on the *Forbes* list in 1987 and 1992 for Indonesia are classified as politically connected by us because they benefited substantially from their ties to the corrupt Suharto regime. As Lambsdorff (2002) notes, in the 1990s businesses run by cronies and relatives of Suharto were granted monopoly rights to control various parts of the economy—what the World Bank euphemistically described as the emergence of conglomerates that seek "to capture rents created by policy-created market distortions." Although there are a handful of politically unconnected individuals who show up on the billionaire list for the first time in 1996, politically connected individuals continue to dominate the list and politically connected wealth inequality peaks at over 7 percent of GDP in 1996 on the eve of the Asian financial crisis. The crisis and the subsequent ouster of President Suharto in 1998 resets the scene both politically and economically as it were. The individuals we classify as politically connected drop out of the *Forbes* list for 2002 and 2007 altogether. Although there is a slight reversal in their fortunes and some of them re-emerge in 2012, even then politically connected wealth inequality clocks in at less than 0.6 percent of GDP—less than a tenth of what it had been just about 15 years earlier. Although overall wealth inequality rebounds after the Asian financial crisis and the global financial crisis to 4.5 percent of GDP by 2012, politically connected wealth inequality never recovers from the shock of the Asian financial crisis and the departure of Suharto from the political scene. In parallel—and in line with our thesis—Indonesia's V-Dem scores jump from about 0.05 until 1996 to about 0.5 in 2002 and remain relatively stable during the rest of our sample period until 2017. Our interpretation of these events is that as the previous economic elites connected to Suharto were displaced by outsider economic elites, Indonesia democratized and it did not revert to autocracy.

In contrast to both the United States and Indonesia, Russia offers yet another flavor of the relationship between wealth inequality and democracy. Russia's development includes two distinct classes or cohorts of politically connected billionaires. The first wave of such individuals who join the *Forbes* list in 2002, include individuals like Mikhail Khodorkovsky of Yukos and Roman Abramovich of Gazprom Neft (formerly Sibneft), who were allies and cronies of Boris Yeltsin. Black et al. (2000) describe it well when they write that Russia ended up "selling control of its largest enterprises cheaply to crooks, who transferred their skimming talents to the enterprises they acquired, and used their wealth to further corrupt the government and block reforms that might constrain their actions." By 2012, several of the billionaires joining the *Forbes* list are those with ties to Vladimir Putin. Hence over the entire time period we study, politically connected wealth inequality is high, with the proportion of billionaire wealth that is politically connected measuring 75 percent or higher—among the very highest in the world. In line with those facts, V-Dem scores also stay low for Russia. Thus even though there is a change in the political guard in Russia, unlike Indonesia, the transition does not bring about democracy, and in fact, V-Dem and Polity scores all decline in the 21st century. As it turns out, rather than a new outsider economic elite challenging the incumbents, a new class of politically connected elite appears on the scene who owe their fortunes to Mr. Putin and have strong incentives to support the autocrat. One can think of Russia as an extreme case of political capitalism, where the political elite helped the economic elite through generous awards of state contracts or sham privatizations and the economic elite returned those favors to their benefactors in politics (Holcombe, 2018).

3 Data-measures of democracy, wealth inequality, and control variables

Before proceeding with our empirical analysis, we define our variables and justify their inclusion. We begin by introducing the three measures of democracy we use in this paper, namely Polity, V-Dem, and the continuous Machine Learning index constructed in Gründler and Krieger (2021). We then describe our measure of wealth inequality, including the subcategories of inherited and politically connected wealth inequality. Finally, we describe the measure of capital mobility and justify its inclusion.

3.1 Measures of democracy

We use Polity scores as our baseline measure of democracy. Polity sums a country's authoritarian and democratic characteristics, creating a −10 to 10 scale, where −10 is fully authoritarian and 10 is fully democratic (Marshall et al., 2017). As shown in Table 1, Polity scores have a mean of 3.93; these scores indicate that the mean country is what the Polity dataset terms an "anocracy," which means that the average country is neither fully democratic nor fully autocratic. We note that Polity only looks at sovereign nations with populations over 500,000, which excludes small countries such as Iceland.

Polity is a broad measure of democracy because it sums a wide variety of both authoritarian and democratic regime characteristics rather than being focused on only a few key dimensions of democracy. Hence, changes in Polity may reflect changes in other political institutions rather than the bedrock institutions of democracy, namely free and fair elections and freedom of expression. Although Polity is the most widely cited democracy index (Vaccaro, 2021), Polity is not a perfect measure of democracy. As Gründler and Krieger (2021) note, Polity's choice of assigning a value of 0 to occupations and transitional governments can generate spurious changes in regime type. Moreover, Polity scores in the middle of the distribution could be achieved by a variety of different combinations of regime characteristics. Thus, if inequality is associated with changes in some characteristics but not others, we may not be able to see that relationship when using Polity scores, which sums all characteristics with equal weighting using a simple additive aggregation technique.[9] Finally, additive rules have a tendency to assign implausibly high values of democracy to autocratic regimes (Gründler & Krieger, 2022).

Setting aside these valid criticisms, we still use Polity as our baseline because of its broad coverage and because of its widespread use throughout the political science and political economy literature. Moreover, if democratic erosion is happening slowly, broad measures may be better able to capture these slow declines by aggregating over more regime characteristics.[10] However, we do not ignore the flaws of Polity scores and choose to use two additional measures of democracy to test the robustness of our results.

Because Polity scores are only one way of measuring a complex, multidimensional concept, we also estimate the exact same regressions with five V-Dem indices on the left-hand side. In addition to the baseline Electoral Democracy index, which captures the idea of "polyarchy" outlined in Dahl (1972), we also use the Liberal, Participatory, Deliberative, and Egalitarian Democracy indices. These latter indices are defined as the weighted sum of the Electoral Democracy index and an additional dimension of democracy (Coppedge et al., 2022b). The Electoral Democracy index captures the extent to which citizens can choose their leaders via elections.[11] The Liberal Democracy index captures the extent to which minority rights are protected from the majority and state repression via constitutional protections, rule of law, and limits on the executive (Coppedge et al., 2022c). The

[9] These regime characteristics can also be redundant, which can artificially increase or decrease a country's Polity score relative to a hypothetical "true" level of democracy.

[10] On the other hand, many theories of the relationship between inequality and democracy posit that the links are between inequality and particular aspects of democracy, for instance, the ability of the poor to vote for redistribution. In this case, using broad measures of democracy may mask the "true" impact of inequality on democracy by mixing different concepts of democracy together (Knutsen & Dahlum, 2022).

[11] More specifically, the Electoral Democracy index measures the foundational aspects of democracy, such as freedom of association and expression, voting rights, free and fair elections, and elections for the executive (Coppedge et al., 2022b).

Table 1 Summary statistics

Variable	Mean	SD	Min.	Max.	N
Polity year	3.931	6.366	−10	10	758
V-Dem electoral democracy	0.538	0.268	0.014	0.926	758
V-Dem liberal democracy	0.428	0.275	0.017	0.896	758
V-Dem participatory democracy	0.353	0.210	0.014	0.805	758
V-Dem deliberative democracy	0.438	0.261	0.008	0.889	758
V-Dem egalitarian democracy	0.411	0.250	0.033	0.880	758
Continuous machine learning index	0.671	0.359	0.000	1	758
Overall billionaire wealth/GDP	1.610	3.903	0	38.815	758
Inherited billionaire wealth/GDP	0.761	1.988	0	16.388	758
Politically connected billionaire wealth/GDP	0.470	2.298	0	38.815	758
Top percentile wealth share	0.305	0.087	0.121	0.572	567
Top decile wealth share	0.638	0.082	0.420	0.889	567
Oil rents	3.757	9.004	0	59.846	758
UN education index	0.545	0.198	0.070	0.933	758
Log of GDP/capita	8.793	1.255	5.821	11.982	758
Capital account openness (KAOPEN)	0.491	0.371	0	1	758
Years of democracy	23.153	32.578	0	208	758
Neighboring average	7.477	4.065	0	14	758
Tax/GDP	16.991	6.723	0.086	48.563	447
Growth of real GDP per capita	0.033	0.076	−0.237	1.008	758

Where available, data are for the 758 observations in the baseline regression

Participatory Democracy index measures the prevalence of electoral and non-electoral elements of direct democracy, including ballot initiatives and civil society. The Deliberative Democracy index captures the extent to which political dialogue is "respectful and reason-based" rather than driven by emotion or coercion, whereas the Egalitarian Democracy index measures the extent to which "inequalities of health, education, and income" inhibit the equal enjoyment of the exercise of political rights (Coppedge et al., 2022c).[12]

In contrast to the more categorical measure of democracy used by Polity, the V-Dem indices are continuous measures on a scale of 0 to 1 that accordingly permit a wider range of outcomes.[13] For example, while the United States has a Polity score of 10 in all years except for 2017, where it falls to 8, its V-Dem Electoral Democracy index value is 0.873 in 1992, 0.872 in 1996, 0.832 in 2002, 0.898 in 2007, 0.902 in 2012, and 0.834 in 2017. Thus, there is considerably more within-country variation for V-Dem than there is for Polity for the United States. As can be seen in Table 1, the mean Electoral Democracy index score is

[12] We find it important to note that the Egalitarian Democracy index is not mechanically related to our measure of wealth inequality. The egalitarian component of the index consists of three sub-indexes. The equal protection index measures the extent to which the law equally protects citizens across social groups and classes. The equal access index measures the *de facto* ability of all people to actively participate in government. The equal distribution of resources index measures how government welfare and infrastructure expenditures, education, and healthcare are distributed in society (Coppedge et al., 2022b). No component of the Egalitarian Democracy index measures the distribution of privately owned assets or wealth.

[13] V-Dem's conceptualization of democracy is also narrower than Polity's. It includes fewer components and hence is less likely to overlap with other institutional outcomes, such as the rule of law or corruption.

0.538, indicating that the average country is neither fully democratic nor fully autocratic. The mean for the other indices is lower, ranging from 0.353 for Participatory Democracy to 0.438 for Deliberative Democracy. V-Dem, rather than using a strictly additive aggregation method, creates its Electoral Democracy index by generating an additive and multiplicative democracy score, and then generates the final index value by taking the arithmetic mean of the additive and multiplicative components. However, both aggregation methods are arbitrary and merely using their arithmetic mean, itself an ad-hoc choice, does not overcome their shortcomings (Gründler & Krieger, 2022). V-Dem's major advantage over other measures of democracy is in the strength of its conceptualization, the wide availability of its raw data, and the detail of its codebook (Gründler & Krieger, 2021).[14] Although V-Dem covers more countries than Polity, we only use those countries that are covered by both Polity and V-Dem to ensure that differences in results are not driven by differences in the underlying sample of countries included in the estimation.

As our final measure of democracy, we use the continuous Machine Learning index of democracy developed in Gründler and Krieger (2021), which uses expert-coded and objective components of democracy covering the areas of political participation, political competition, and freedom of opinion. Political participation looks at *de jure* and *de facto* restrictions on voting as well as voter turnout, political competition measures party pluralism and electoral competitiveness, and freedom of opinion measures the ability of citizens to express political views, even when those political opinions go against the prevailing government (Gründler & Krieger, 2021). Rather than rely on simplistic aggregation techniques, the authors use Support Vector Machines to generate a flexible way of generating democracy score values. These scores range from 0 to 1, and, in contrast to V-Dem scores, include more values above 0.9 and below 0.1; this is because additive indexes tend to assign values to democracy (autocracy) which are too low (too high) (Gründler & Krieger, 2022). Lastly, the Machine Learning index has the advantage of being more sensitive to democratic transitions than are additive indexes like Polity because the Machine Learning index registers larger changes as regime characteristics change. While the Machine Learning index has many points in its favor, we do not rely on it alone.

We use all three measures of democracy rather than preferring one over the other, because although they each have a pairwise correlation coefficient above 0.87, as suggested by Fig. 1, the measures are fundamentally different. Not only are the V-Dem indices continuous as opposed to the Polity scores which are categorical, the V-Dem indices also allow us to examine multiple conceptions of democracy separately. On the other hand, Polity looks at the sum of democratic or authoritarian characteristics.[15]

Lastly, the Machine Learning index has the advantage of not relying on a simplistic and arbitrary aggregation rule, but instead has a flexible method of aggregation that allows for more accuracy in assigning values of democracy to the tails of the distribution, which in turn allows it to be more sensitive to democratic transitions than strictly additive measures like Polity. The choice of democracy measure meaningfully impacts regression results (Cheibub et al., 2010; Gründler & Krieger, 2022), especially with regard to variables related to modernization, such as GDP per capita and education (Högström, 2013). Relevant for our paper, recent research has found that because democracy measures have diverged since the 1980s, the choice of democracy index can impact our coefficient

[14] For the codebook and methodology, see Coppedge et al. (2022b, 2022c).

[15] Polity and V-Dem scores are generated using different processes. While Polity uses in-house experts that code all countries using country-specific reports, V-Dem uses observational data, expert surveys, and in-house experts (Skaaning, 2018).

estimates and conclusions given our sample dates (Vaccaro, 2021). Thus, because there is no one theoretically best democracy index for our particular research question, we prefer to use multiple measures to show that our results are robust.

Although some scholars argue for a dichotomous rather than a continuous measure of democracy (Cheibub et al., 2010), we use continuous measures in part because we are interested in the gradual erosion of democratic norms that has marked the recent wave of autocratization (Lührmann & Lindberg, 2019). The use of a continuous measure of democracy also enables us to look at the decoupling of rights and democracy noticed in Mounk (2018). To the extent that we are seeing the rise of "democracy without rights" we may see a decline in democracy scores that include rights to free expression without those declines translating into transitions to autocracy if core rights to political participation remain unaffected. Moreover, because dichotomous indices cannot capture the gradual nature of most democratic transitions, they sometimes assign the year of transition too late (Gründler & Krieger, 2022). Given the specifics of this recent wave of autocratization, we are particularly concerned with the possibility of dichotomous measures mislabeling the date of transitions. Lastly, a focus on dichotomous measures of democracy means that we would be unable to see the impact of inequality on the "quality" of democracy. Because continuous measures view democracy as a matter of degree, by definition they are able to account for small variations in the level of democracy and can more accurately track the gradual changes that may accumulate into democratic transitions or reversals (Coppedge et al., 2022a).

To provide a concrete example of what we mean by the "quality" of democracy, Albertus and Menaldo (2018) argue that many autocracies transition into democracies whose constitutions and institutions are designed to benefit the former political incumbents and their economic allies. While these countries are democratic, they do not reflect the policy preferences of the median voter and are biased more towards the preferences of the economic,[16] and political elites.[17] Thus, focusing on a dichotomous measure of democracy would miss the rich variation in the levels of democracy within and across countries and the institutional rules that elites may prefer which, while anti-majoritarian, are not undemocratic.

3.2 Billionaire wealth as a measure of wealth inequality

Data for billionaire wealth comes from Bagchi et al. (2019), which extends the work in Bagchi and Svejnar (2015). The latter paper uses the *Forbes* list of billionaires to create a measure of wealth inequality for a large cross-section of countries. Billionaire wealth as a share of GDP is a valid measure of inequality due to correlation coefficients of roughly 0.5 with multiple measures of wealth inequality, as noted in Bagchi and Svejnar (2015), and it also captures the increase in inequality over time noted by Piketty (2014). Moreover,

[16] Restrictions on suffrage or institutional rules that make voting onerous are one such potential mechanism. To the extent that increased voter turnout increases top marginal tax rates (Sabet, 2023) economic elites may have incentives to lobby for rules that make voting more difficult in order to protect their incomes.

[17] Myanmar's 2008 constitution provides a great example of this phenomenon. As Nehru (2015) notes, it included several provisions to ensure that the reins of power remained firmly in the hands of the military, chief among them being "Article 436 that gives the military one-quarter of the seats in the upper and lower houses of the national parliament and one-third of the seats in the state/regional parliaments. In addition, because constitutional amendments must receive more than 75 percent of the vote in parliament, the military's mandated 25 percent presence gives it effective veto power over any proposed changes."

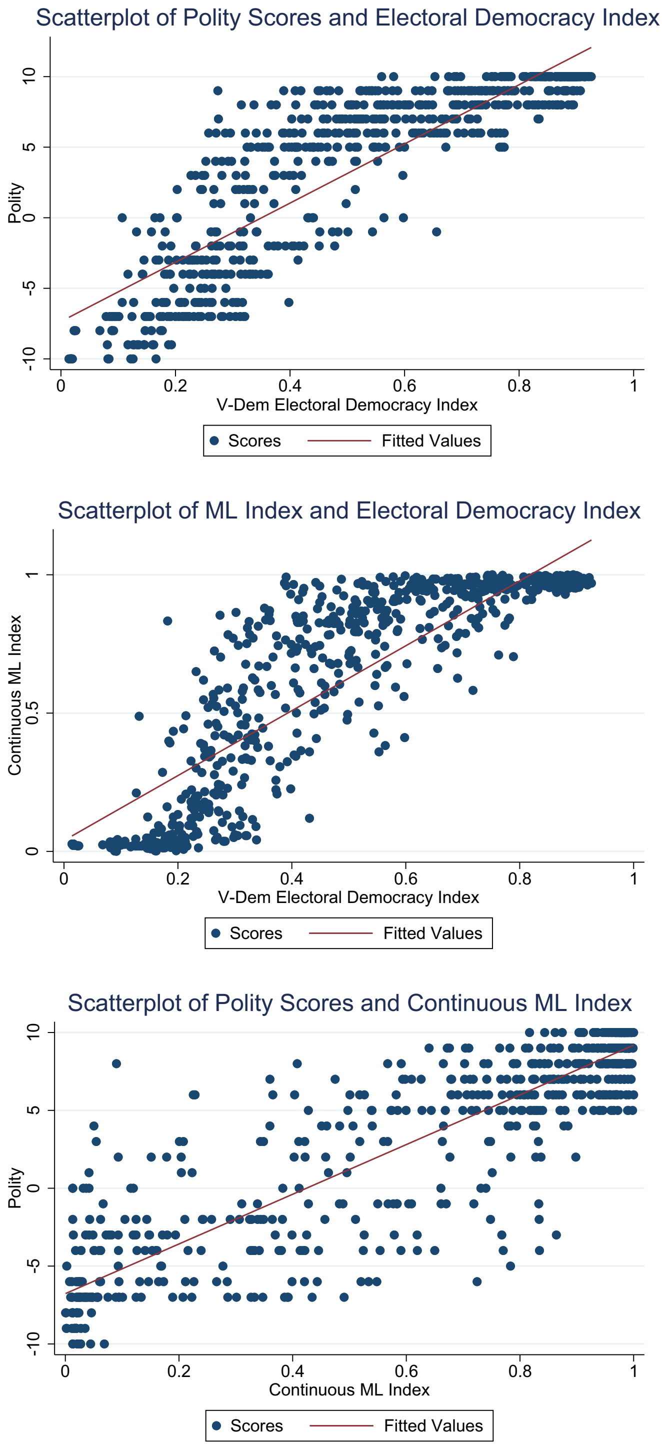


Fig. 1 Scatterplots of different democracy measures in the same year

this measure is appropriate for this particular research question, as we explained at length above. A complete list of countries with at least one billionaire can be found in Table 11.

To maximize our sample size, we impute a value of 0 for countries that do not have billionaires.[18] The mean of 1.6 indicates billionaires tend to hold wealth worth around 1.6 percent of GDP on average, but this figure is influenced by the number of countries without billionaires. When looking only at countries that have billionaires, the mean rises to 3.9, or roughly 4 percent of GDP. In the data, Georgia (2012) exhibits the highest level of wealth inequality at 38.8 percent of GDP. Limiting ourselves to countries with populations above 10 million, the Philippines (1996) has the highest level of wealth inequality at 25.1 percent of GDP.

As described more fully below, we classify billionaires based on whether they were self-made or inherited, and separately as politically connected or unconnected. The maximum for inherited wealth inequality is 16.3 percent of GDP in Bahrain in 1996, while the maximum for politically connected wealth inequality is for Georgia in 2012 at 38.8 percent of GDP. Limiting ourselves to countries with populations above 10 million, Russia (2007) has the highest level of politically connected wealth inequality at 17.0 percent of GDP. These values indicate that at least in some countries both inherited and politically connected billionaires control large amounts of national wealth.[19]

Of the 56 countries with billionaires in at least one time period, 44 have at least one billionaire who inherited their wealth, whereas 37 have at least one politically connected billionaire. The countries with politically connected billionaires range from more obvious suspects such as Russia or Malaysia, where natural resource billionaires exploited connections to political elites, to countries such as Italy, where media magnates and car manufacturers used connections to politicians to get ahead. Also included in the category of politically connected billionaires are the individuals at the helm of South Korean business conglomerates, the Chaebols, such as Mr. Lee Kun-Hee of Samsung who had extensive connections to the government.

To determine whether a billionaire inherited their wealth or is politically connected, we follow the methodology described in Bagchi and Svejnar (2015) and Bagchi and Svejnar (2016), which uses LexisNexis and ProQuest to search newspaper articles for mentions of inheriting a business or for evidence of corruption and connections, respectively. The key question we ask ourselves when classifying an individual as politically connected is the following: Absent the political connections that this individual possesses, would this person have been as affluent as s/he is today? Examples of events that would lead us to classify an individual as politically connected are privatization deals run in a noncompetitive manner that provide the start to someone's business career or the grant of an exclusive license / concession to an individual as part of a quid pro quo kind of arrangement with politicians.[20] The privatizations undertaken in Boris Yeltsin's Russia soon after the fall of Communism offer a great example of events that led to the emergence of politically connected billionaires in that country. In fact, while the *Forbes* list does not list a single individual as a billionaire in 1996, seven individuals are listed in 2002 and all of those billionaires benefited from the sham privatization exercises conducted under President Boris Yeltsin.

[18] Our results for overall wealth inequality are robust to dropping these countries.

[19] Although we operationalize wealth inequality by normalizing billionaire wealth by GDP throughout the paper, we also present results obtained by normalizing billionaire wealth by population and the country's capital stock in Online Appendix Tables A.7 and A.8.

[20] A full classification of billionaires into the two categories of politically connected and politically unconnected is available from the authors on request.

These include individuals like Mikhail Khodorkovsky of Yukos, Roman Abramovich of Gazprom Neft (formerly Sibneft), and Vladimir Potanin of Norilsk Nickel. Take Mr. Potanin for example. A former deputy prime minister under President Yeltsin, he took over the assets (but not the debts) of two Soviet-era foreign trade banks in the early nineties and subsequently bought "control of metals giant Norilsk Nickel and oil company Sidanko in lopsided privatization auctions in 1995."[21] As a story on PBS additionally notes:[22] "Later, Potanin became one of the principal authors of the Loans for Shares program, in which the Russian government traded ownership in state industries for loans. The controversial program was administered through auctions, but only select bidders were invited to attend, usually at the discretion of President Boris Yeltsin's daughter. Potanin was among the invited bidders." Per that analysis, we classify Mr. Potanin as politically connected because his connections were indispensable to his success as a businessman.

Mr. Liem Sioe Liong of Indonesia provides us with another example of a politically connected billionaire, but in this instance because he benefited from the grant of an exclusive license to engage in the import of a good, stemming from his proximity to President Suharto. As his 1987 *Forbes* profile notes, Mr. Liong in the 1960s "developed close ties with an ambitious young lieutenant colonel named Suharto. By the time Suharto came to power in 1965, he and Liem were already business partners. In 1968 Suharto granted one of Liem's companies a monopoly for importing cloves. Monopolies in flour and cement, as well as open credit lines from state banks, soon followed." Likewise, a Washington Post article notes:[23] "Bogasari Flour Mills was formed by Liem in 1969 with Suharto's foster brother, Sudwikatmono, as president-director and a 4 percent shareholder, according to an official company registration. The firm was awarded a license for *exclusive* flour-milling and marketing rights in western Indonesia, accounting for about 80 percent of domestic consumption and worth about $240 million in annual sales. By 1982 Bogasari had taken over the management of a company awarded a license for the eastern region, giving it *virtual monopoly control* of the Indonesian flour market. In 1968, Liem and a stepbrother of Suharto, Probosutedjo, were granted *exclusive rights* to import cloves, a particularly lucrative concession because of their use in Indonesian cigarettes." News articles like these help substantiate our claims that Mr. Liong benefited materially from political connections to President Suharto and that leads us to classify him as politically connected.[24]

It is also important for us to point out that we do not classify an individual as politically connected simply because they have been involved in politics themselves or if they have made large campaign contributions, absent further evidence that such involvement in politics or campaign contributions led to quid pro quo favors from those in government. As an example of a politician that we do not classify as politically connected, we offer Mr. JB Pritzker, the current Democratic Governor of Illinois serving his second term in office.

---

21 https://www.forbes.com/billionaires2002/LIRKZ32.html.

22 https://www.pbs.org/frontlineworld/stories/moscow/potanin.html .

23 https://www.washingtonpost.com/archive/politics/1986/04/27/corruption-issue-could-cloud-suhartos-impressive-record/49e076d5-a764-464e-b314-1f212a06b352/.

24 Fisman (2001) attempts to quantify the value of political connections and finds Indonesia to be especially fertile territory. It obtains estimates of the value of such connections by exploiting a string of rumors about President Suharto's health during his last few years in office. The paper scores the companies affiliated with "longtime Suharto allies", the Salim Group run by Liem Sioe Liong and the Barito Pacific Group run by Prajogo Pangestu, as five on a scale of five—the *highest* score that it also assigns to companies associated with President Suharto's children. Thus our classifications of Indonesian billionaires, while undertaken independently, are consistent with those in a very well-cited paper that examines political connections.

Although he is clearly a politician, Mr. Pritzker's wealth stems from his family's part ownership of the hotel chain Hyatt and *not* from his political connections. Likewise, take Pete Ricketts, the former two-term Republican Governor of Nebraska (2015–2023) and the current junior Senator from the state. Mr. Pete Ricketts is the son of Joe Ricketts, the founder of TD Ameritrade, and the senior Mr. Ricketts' wealth is estimated as exceeding $3 billion as of March 2023. Nonetheless, we do not classify Mr. Joe Ricketts as politically connected because the success of TD Ameritrade, on which his fortune is based, is unrelated to his son's career as a politician.

We apply the same underlying principle as far as campaign contributions go. Some of the most prolific contributors in U.S. politics over the last decade have been Mr. Sheldon Adelson (with donations exclusively to Republicans), Mr. Michael Bloomberg (donating predominantly, but not exclusively, to Democrats), and Mr. George Soros (donating exclusively to Democrats). For example, as a report by the Center for Responsive Politics on the eve of the 2020 election noted:[25] "The Adelsons (Sheldon and Miriam Adelson) have been among the top donors for the past decade. They gave more than anyone else in 2018 and 2012, and were just short of the No. 1 spot in 2016. In the past 10 years, they've given a total of $480 million, typically made in his-and-hers political contributions, each donating the same amount to candidates and groups." In 2020, the couple set a new record for donations from individuals in a single election cycle, contributing a total of over $218 million. If one were to define political connections based on one's involvement with political campaigns and contributions to those, one would almost surely classify Mr. and Mrs. Adelson as politically connected. Yet even a critic of Mr. Adelson would be hard-pressed to attribute his rise to riches to his political connections. For instance, The New York Times—hardly an outlet sympathetic to Mr. Adelson—noted in his obituary that Mr. Adelson's rise began in 1979 when he and four partners started a Las Vegas computer trade show called Comdex that drew sellers and buyers to see the latest technology in a field that was growing exponentially and ended up being "the nation's top computer exhibition." Likewise, with Mr. Bloomberg and Mr. Soros, their wealth can be attributed to the success of their commercial ventures, Bloomberg L.P. and Soros Fund Management LLC, and not to any ties or connections that they may have developed to politicians during the course of their business or philanthropic activities.

The reasonableness of our approach to classifying billionaires as politically connected or unconnected can be gleaned from the following set of facts: For the four countries with billionaires in all 6 years and no politically connected billionaires (Canada, Netherlands, Sweden, and Switzerland), their median rank on Transparency International's Corruption Perception Index was 6 out of 180 countries in 2019,[26] whereas the five countries with the highest levels of politically connected billionaire wealth as a percentage of GDP (Georgia, Kazakhstan, Lebanon, Russia, and Ukraine) had a median rank of 126 out of 180 countries.[27] Additionally, we also examine the degree to which our

[25] https://www.opensecrets.org/news/2020/10/adelsons-set-new-donation-record.

[26] A higher rank on the Corruption Perceptions Index indicates that a country is perceived as being more corrupt by experts knowledgeable about the country. In the 2019 rankings by Transparency International, Denmark and New Zealand shared the 1st spot and were viewed as the least corrupt countries in the world whereas Yemen, Syria, South Sudan, and Somalia were perceived as the most corrupt countries in the world.

[27] This list is calculated using the average of politically connected wealth inequality in years where these countries had at least one billionaire. These results are comparable to those found in Bagchi and Svejnar (2015) (refer Table A2, p. 528).

measure of the importance of politically connected wealth correlates with the measure of political connections developed independently in Faccio (2006) wherein the author examines whether controlling shareholders and top officers are connected with national parliaments or are members of their governments. For the 42 countries listed in Table 2 of Faccio (2006) that also have billionaires at least once in our sample period, we find that politically connected wealth as a share of GDP is correlated with the proportion of firms in those countries that are politically connected with a correlation coefficient of 0.662 (p-value < 0.001) and with the market capitalization of connected firms as a percent of overall market capitalization with a correlation coefficient of 0.735 (p-value < 0.001),[28] Additional evidence confirming the validity of our measure of politically connected wealth inequality is provided in Bagchi and Svejnar (2015) and we refer the interested reader to Sect. 4.2 of the paper. Our discussion in that section of the paper shows that we obtain positive and statistically significant coefficients by regressing politically connected wealth inequality against (1) the ICRG Corruption Score, as well as the (2) Instrumental Variable developed in Easterly (2007) - the exogenous suitability of land for wheat versus sugar (LWHEATSUGAR).

One final note is that although the *Forbes* list is a fairly comprehensive measure of billionaire wealth, it does not include monarchs or dictators, some of whom are rumored to own or actively control vast financial resources. Thus, we recognize that our measure of wealth inequality may be systematically lower in autocracies and monarchies if the wealth of royals and oligarchs in politics is systematically under-counted.[29] We attempt to address this concern by dropping from our sample countries that had their heads of state ever chosen by hereditary succession during our sample period (as is true for Japan, Thailand, and Saudi Arabia for example) or a royal council (as is true for Malaysia and the United Arab Emirates for example) and then re-estimating our regressions. We provide the results from that exercise in Online Appendix Table A.5 and they line up very closely to the results we report in our baseline specification in Table 3 of the paper. Likewise, we also present results obtained by excluding both autocracies and monarchies where wealth inequality may be the most understated in our Online Appendix Table A.6, and find the results similar to those reported in Table 3.

3.3 Capital mobility and the interaction with wealth inequality

According to Boix (2003), as the assets held by economic elites become more vulnerable to expropriation by voters demanding redistribution, the elites' response will be to oppose democracy, because democracy becomes increasingly costly if they are unable to move their assets to safety. However, economic elites may also be fearful of expropriation by

[28] Furthermore, in a univariate regression between politically connected wealth as a share of GDP and either the proportion of firms that are politically connected or the fraction of market capitalization represented by politically connected firms, the regression coefficients are significant at the 5% level (or higher) and additionally, the values of R-squared, with *just a single variable included,* exceed 0.40. Those additional regressions provide further assurance that our measure of politically connected wealth inequality is reasonable as it lines up well with measures in Faccio (2006)—a well-cited paper that arrives at political connections using an entirely different approach.

[29] We note though that in many constitutional monarchies, the monarch or other royals are often legally unable to put the institutional wealth of the crown for personal use. For example, in the United Kingdom the Crown Estate manages the property of the King but they note that *"it is not the private property of the monarch*—it cannot be sold by the monarch, nor do revenues from it belong to the monarch."

political elites, and thus may prefer democracy because it more reliably protects property rights than authoritarian regimes (Treisman, 2020). If this theory holds, countries with high capital account openness will see a smaller impact from wealth inequality because their wealth is less vulnerable to being either redistributed by voters or expropriated by political elites. Because the effect of wealth inequality on democracy is ambiguous, we simply hypothesize that the impact of the interaction between capital account openness and wealth inequality will be opposite that of wealth inequality, and will thus reduce the net impact towards 0 as capital account openness increases.

We measure capital mobility using the Chinn-Ito index (Chinn & Ito, 2006), which is widely used in the literature on financial systems (Baltagi et al., 2009; Ilzetzki et al., 2019), and in prior work examining the impact of income inequality on democracy (Policardo & Carrera, 2020). We follow the standard practice of referring to this index as KAOPEN. The Chinn-Ito index, ranging from 0 to 1, measures the number of restrictions on international financial transactions. Although values of the index are available on an annual basis from 1970 to 2017, as with our other independent variables and controls, we are interested in the values of the index for 1987, 1992, 1996, 2002, 2007, and 2012.[30] While we include KAOPEN on its own, we are primarily interested in its interaction with wealth inequality to test the theory in Boix (2003).

4 Empirical approach

The aim of this study is to test the relationship between wealth inequality, measured as billionaire wealth as a percentage of GDP, and levels of democracy using Polity, the Varieties of Democracy project, and the Machine Learning index. Due to the persistence of prior regime type, we use five-year gaps between observations, and cover the years 1992, 1996, 2002, 2007, 2012, and 2017.[31] Because the democratizing impacts of economic development take time to develop (Treisman, 2020), we lag our independent variables, with the exception of years of democracy. The use of a lag is important with regard to inequality specifically because there is evidence that democracy, and in particular transitions to democracy, reduce asset values via an increased risk of redistribution (Miller, 2023). Thus, higher levels of democracy will reduce the numerator of our measure of inequality by lowering the value of assets held by billionaires. Moreover, because democracy promotes economic growth (Acemoglu et al., 2019), democratization also increases the denominator of our measure of inequality. Democracy therefore will serve to decrease our measure of inequality mechanically, without even considering any reduction in inequality that may be

[30] The Kernel Density of KAOPEN can be found in the bottom panel of Fig. 2.

[31] The choice of 1996 is motivated by a change in the way *Forbes* covered billionaire wealth in 1997. As Bagchi and Svejnar (2015) explains: "*Forbes* magazine changed its editorial policy for four years, between 1997 and 2000. In these years, they included only those billionaires who were either self-made (e.g., Warren Buffett) or those who inherited their wealth and were actively managing it themselves (e.g., Carlos Slim Helu of Mexico). This leads to the exclusion of billionaires from around the world who simply inherited their wealth and were no longer actively involved themselves in growing their businesses, such as the duPonts and Rockefellers in the U.S. [...] Given this limitation of the 1997 list, we use the 1996 list instead." Because we are interested in how inherited wealth inequality impacts democracy, it is important that we include all inherited billionaires and not just those who are actively managing their fortunes.

brought about by the expansion of the franchise (Husted & Kenny, 1997). Thus, because of the endogenous relationship between inequality and democracy, we find it particularly important to lag wealth inequality.

For our baseline results, we use OLS regression, as it is a commonly used empirical technique when working with continuous measures of democracy (Midlarsky, 1992; Muller, 1995). We do not use probit analysis as we are more interested in the level of democracy than in a binary classification of countries into democracies and autocracies. Moreover, because autocratization in recent history has begun to blur the line between autocracy and democracy, and because these breakdowns have been gradual (Lührmann & Lindberg, 2019), econometric analysis on a more continuous scale is justified due to different regime transition dynamics than were prevalent in previous waves of democratization. Because of Nickell (1981) bias, we do not include a lag of the dependent variable. Instead, we capture the effects of persistence through two different channels. In terms of temporal persistence, we look at the number of years a country has been a democracy, and we control for regional persistence by including the average Freedom House scores of neighboring countries. We follow the specification used in Leeson and Dean (2009) and code islands with no land borders as having a zero for neighboring country average, as they have no neighbors.[32] Therefore, the United Kingdom, which shares a border with Ireland, has a neighboring average, as does Indonesia, which shares a border with East Timor, Malaysia, and Papua New Guinea, whereas Japan, lacking a neighbor with which it shares a land border, is assigned a value of zero.

Our regression equation takes the following form:

$$\begin{aligned} Polity_{it} =& \beta_0 + \beta_1\left(\frac{B}{GDP}\right)_{it-1} + \beta_2 KA_{it-1} \\ &+ \beta_3\left(\frac{B}{GDP}\right)_{it-1} * KA_{it-1} + \mathbf{B}_4\mathbf{C}_{it-1} + \alpha_i + \nu_t + \varepsilon_{it} \end{aligned} \tag{1}$$

where i indexes country and t indexes year. B/GDP is billionaire wealth as a percentage of GDP, KA is capital account openness, $\mathbf{C}$ is the vector of controls,[33] α_i are country fixed effects, ν_t are period fixed effects, and ε_{it} is the error term. We use fixed effects instead of random effects, as the Hausman test finds random effects to be inconsistent.[34] One other advantage of the use of fixed effects is that it allows us to control for time-invariant factors such as cultural influences (Woodberry, 2012; Gorodnichenko & Roland, 2021), ethnic fractionalization (Leeson, 2005), and colonial status (Bernhard et al., 2004), without including them in the regression model. We also repeat this regression with inherited or politically connected wealth inequality and the other measures of democracy.

[32] As can be seen from Online Appendix Table A.2, our results are robust to dropping such countries.

[33] Section 1 in the Online Appendix lists all of the controls, including their justification for inclusion.

[34] However, we also confirm that all our results are robust to the use of random effects specifications.

5 Results

5.1 Results obtained without interacting inequality with capital mobility

Table 2 displays the results we obtain when excluding an interaction term between wealth inequality and capital mobility, while still including capital mobility as a control variable. In this table and our subsequent tables, unless otherwise specified, results for overall wealth inequality are in Panel A, those for inherited wealth inequality in Panel B, and results for politically connected wealth inequality in Panel C. Our results in Table 2 confirm the findings in Houle (2009) that there is no statistically significant relationship between inequality and democracy. Moreover, this null result holds for all of the measures of democracy considered as well as the sub-components of wealth inequality. Thus, for the world at large, we find no significant relationship between wealth inequality and democracy.[35]

This null result in terms of the impact of wealth inequality on democracy matches the findings reported in much of the empirical literature. For instance, in a recent paper published in this very journal, Rød et al. (2020) considers as many as 67 proposed determinants of democratization and finds that the only robust factors explaining democratic survival are income and the existence of a law-abiding bureaucracy. For democratization, the authors report finding a positive impact from political protests, being in a democratic neighborhood—a finding echoed in Leeson and Dean (2009) as well—and the global proportion of democracies. Notably, the authors do not find a robust effect in terms of the impact of inequality on democratization. Likewise, Ahlquist and Wibbels (2012) summarize the literature by noting that many papers "have failed to find any relationship at all between various measures of inequality and regime type"and speculate that "we suspect these conflicting findings result in part from the difficulty of identifying the causal relationship between inequality and democracy." In describing their own findings, the authors note that in spite of assembling "a large dataset extending back to the last third of the nineteenth century," they "find no evidence for a systematic link between world trade and democracy in any direction, nor do we observe a link between inequality and democratization."

5.2 Results obtained interacting inequality with capital mobility—baseline specification

Consistent with our discussions earlier that capital mobility might moderate the impact of wealth inequality on democracy, we now provide results from our baseline specification which includes an interaction between wealth inequality and capital account openness. Our results in Table 3 indicate that although there is a relationship between wealth inequality and democracy, these results are not robust to the choice of measure of democracy. As can be seen in columns (2) through (7), overall wealth inequality is significantly associated only with declines in Liberal and Deliberative Democracy, as well as the continuous Machine Learning index of democracy. Although the lack of robustness is contrary to the theory in Boix (2003), they do line up with the larger literature in finding that there is little relationship between wealth inequality and democracy in the world at large. Likewise, our

[35] These null results hold when we limit our sample to only those countries that have had billionaires at least once. See Online Appendix Table A.1.

Table 2 Effects of wealth inequality and capital account openness on democracy without an interaction term

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: Wealth as percent of GDP							
Wealth Inequality	0.00331	0.000206	−0.000271	0.0000273	0.000118	0.000205	0.000627
	(0.0197)	(0.00133)	(0.00136)	(0.000989)	(0.00156)	(0.00105)	(0.00229)
KAOPEN	−0.191	−0.0163	−0.0137	−0.00895	−0.0179	−0.0150	0.00606
	(0.736)	(0.0248)	(0.0235)	(0.0177)	(0.0232)	(0.0165)	(0.0477)
R^2	0.096	0.105	0.090	0.144	0.107	0.131	0.147
Panel B: Inherited Wealth/GDP							
Inherited Wealth Inequality	0.0704	0.00121	−0.0000219	0.000448	0.000684	0.000663	0.00463
	(0.0470)	(0.00242)	(0.00263)	(0.00156)	(0.00269)	(0.00180)	(0.00553)
KAOPEN	−0.210	−0.0166	−0.0137	−0.00907	−0.0181	−0.0152	0.00471
	(0.736)	(0.0250)	(0.0237)	(0.0178)	(0.0234)	(0.0167)	(0.0475)
R^2	0.097	0.105	0.090	0.145	0.108	0.131	0.149
Panel C: Pol. Connected Wealth/GDP							
Pol. Connected Wealth Inequality	0.000576	0.00123	0.000776	0.000899	0.00144	0.00118	−0.000607
	(0.0339)	(0.00226)	(0.00231)	(0.00171)	(0.00266)	(0.00173)	(0.00396)
KAOPEN	−0.192	−0.0160	−0.0134	−0.00869	−0.0175	−0.0147	0.00571
	(0.738)	(0.0249)	(0.0236)	(0.0178)	(0.0234)	(0.0167)	(0.0477)
R^2	0.096	0.105	0.090	0.145	0.108	0.132	0.147
For all panels							
N	758	758	758	758	758	758	758
No. of countries	149	149	149	149	149	149	149
Country/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index from Gründler and Krieger (2021). We use the first lag of all independent variables except for years of democracy

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

results in Panel B indicate that inherited wealth inequality has no significant relationship with democracy.

When examining the impact of politically connected wealth inequality in Panel C of Table 3, we see that although there is no significant relationship between wealth inequality and Polity scores, there is a significant, negative relationship between wealth inequality and democracy as measured by the various V-Dem indices or the Machine Learning index. Lastly, the interaction term between politically connected wealth inequality and capital account openness is significant and positive, suggesting that higher levels of capital account openness attenuate the negative impact of politically connected wealth inequality. Our results indicate that for a country with no capital mobility, a one standard deviation increase in politically connected wealth inequality (2.29 percent of GDP) is associated with a 0.009 point decrease in Electoral Democracy and 0.017 point decline in Machine

Table 3 Effects of wealth inequality, capital account openness, and their interaction on democracy—baseline specification

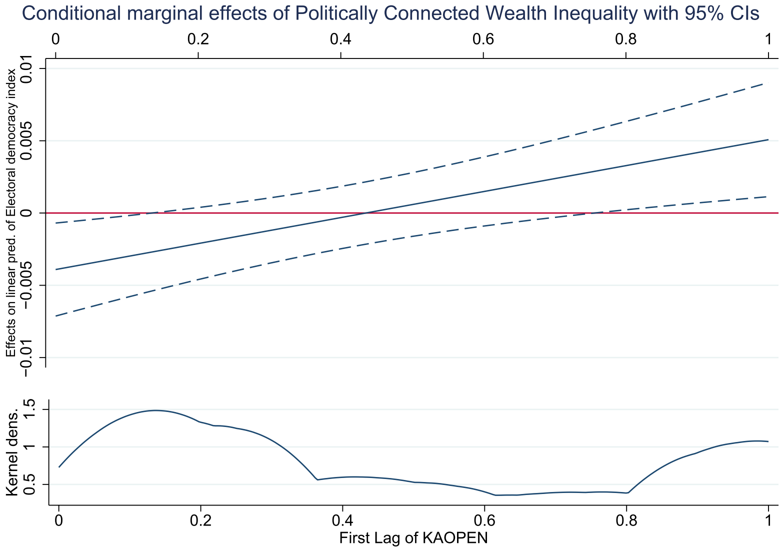
	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: Wealth as percent of GDP							
Wealth inequality	−0.0133	−0.00328	−0.00379*	−0.00217	−0.00489**	−0.00270	−0.00766**
	(0.0470)	(0.00202)	(0.00216)	(0.00137)	(0.00233)	(0.00167)	(0.00345)
KAOPEN	−0.228	−0.0240	−0.0216	−0.0138	−0.0291	−0.0214	−0.0123
	(0.760)	(0.0263)	(0.0248)	(0.0186)	(0.0244)	(0.0171)	(0.0494)
W.Inequality*KAOPEN	0.0258	0.00539*	0.00545*	0.00340	0.00775**	0.00449*	0.0128*
	(0.0617)	(0.00296)	(0.00309)	(0.00214)	(0.00340)	(0.00242)	(0.00684)
R^2	0.096	0.108	0.094	0.147	0.114	0.136	0.152
Panel B: Inherited Wealth/GDP							
Inherited Wealth Inequality	−0.0824	−0.00555	−0.00725	−0.00397	−0.0129	−0.00432	−0.0161
	(0.181)	(0.0107)	(0.0123)	(0.00671)	(0.0115)	(0.00832)	(0.0147)
KAOPEN	−0.334	−0.0221	−0.0195	−0.0127	−0.0292	−0.0193	−0.0121
	(0.764)	(0.0261)	(0.0248)	(0.0186)	(0.0244)	(0.0169)	(0.0500)
W.Inequality*KAOPEN	0.211	0.00935	0.01000	0.00611	0.0188	0.00688	0.0287
	(0.236)	(0.0127)	(0.0145)	(0.00825)	(0.0138)	(0.0100)	(0.0234)
R^2	0.099	0.107	0.092	0.146	0.115	0.133	0.153
Panel C: Pol. Connected Wealth/GDP							
Pol. Connected Wealth Inequality	0.00680	−0.00391**	−0.00434***	−0.00246*	−0.00424**	−0.00329**	−0.00759***
	(0.0733)	(0.00164)	(0.00158)	(0.00125)	(0.00189)	(0.00159)	(0.00271)
KAOPEN	−0.186	−0.0212	−0.0186	−0.0121	−0.0233	−0.0193	−0.00142
	(0.742)	(0.0254)	(0.0241)	(0.0180)	(0.0237)	(0.0168)	(0.0486)
W.Inequality*KAOPEN	−0.0109	0.00899***	0.00893***	0.00587**	0.00993***	0.00781***	0.0122**
	(0.112)	(0.00291)	(0.00300)	(0.00243)	(0.00348)	(0.00233)	(0.00590)
R^2	0.096	0.110	0.095	0.149	0.114	0.140	0.149

Table 3 (continued)

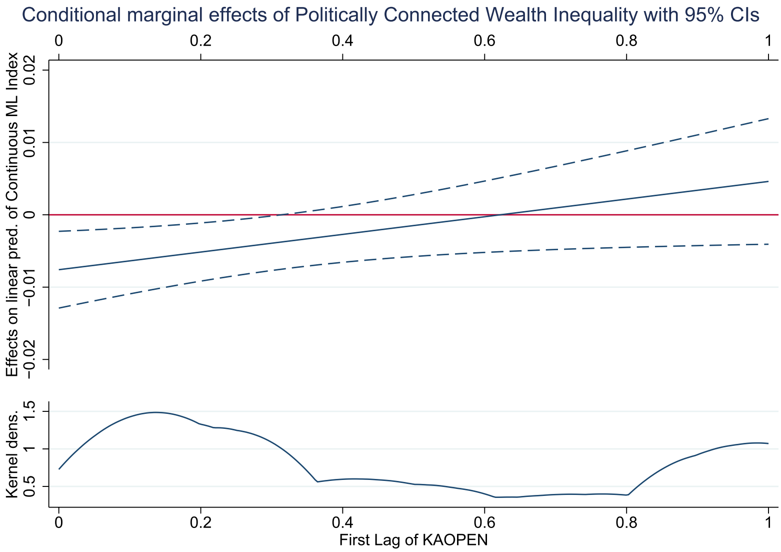
	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
For all panels							
N	758	758	758	758	758	758	758
No. of countries	149	149	149	149	149	149	149
Country/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index. We use the first lag of all independent variables except for years of democracy

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$



(a) Conditional Marginal Effects of Politically Connected Wealth Inequality. This plot corresponds to column (2) of Panel C in Table 3. The intercept of the top panel represents the coefficient on politically connected wealth inequality when capital account openness is at its minimum of 0.



(b) Conditional Marginal Effects of Politically Connected Wealth Inequality. This plot corresponds to column (7) of Panel C in Table 3.

Fig. 2 Marginal effects plots for electoral democracy and machine learning index

Learning Democracy scores, a finding that can be seen in Fig. 2. To give a concrete example, in India the Electoral Democracy index declined from 0.735 in 2007 to 0.704 in 2012. During the preceding five years, politically connected wealth inequality increased by 5.5 percent of GDP. With their capital account openness of 0.166, we find that politically connected wealth inequality explains roughly 43 percent of the decline in democracy in India.[36] By looking at the conditional marginal effects plot in Fig. 2, we can also see that once capital account openness reaches a value of just under 0.2, we can no longer state that the effect of politically connected wealth inequality on the electoral democracy index is significantly different from 0. This figure captures the theory in Boix (2003) that capital mobility is a key factor preventing inequality from significantly damaging democracy. Our results in Table 3 indicate that the type of inequality matters; inequality arising from inheritance is not a significant determinant of democracy, but wealth inequality resulting from the abuse of political connections erodes democracy when capital mobility is low.

Thus, although building on the theory laid out in Boix (2003), we are able to add the important caveat that this relationship is conditional on inequality being generated through political connections rather than through market processes. Our results could either mean that economic elites who enriched themselves through the political process in democratic countries oppose democracy because they believe it endangers their position, or that voters believe that democracy has failed to address the issue of inequality and corruption (Krieckhaus et al., 2014).

5.3 Decomposing wealth inequality into politically connected and unconnected wealth inequality

To better understand the impact of wealth inequality and its components on democracy, in Table 4 we decompose overall wealth inequality into its two components: politically connected and unconnected wealth inequality and interact them both with capital account openness. We examine three samples: all countries in Panel A, democracies in Panel B, and countries with billionaires in Panel C. In Panels A and B, we find a statistically significant, negative relationship between politically connected wealth inequality and all five V-Dem indices. Moreover, in all three panels, we find a negative and statistically significant relationship between politically connected wealth inequality and the Machine Learning index. We also note that our estimates for politically connected wealth inequality are largely unaffected by the inclusion of politically unconnected wealth inequality and, in fact, a formal t-test of differences between the corresponding coefficients in Tables 3 and 4 confirms as much.

In contrast to the robust negative impact of politically connected wealth inequality, we do not find a statistically significant relationship between politically unconnected wealth inequality and democracy in even *one* of the 21 regressions in Table 4, either in the presence or absence of capital account openness. This is a critically important result; it suggests that to the extent some of our prior results suggest a negative relationship between wealth inequality and democracy (in the absence of capital account openness), those effects are driven by the impact of politically connected wealth inequality. All in all, our results from

[36] A 5.5 percentage point increase in wealth inequality implies a decline in V-Dem scores of $5.5 * (-0.00391 + (0.00899 * 0.166)) = -0.0133$ points. A decline in V-Dem scores of 0.031 points implies that $0.0133/0.031 = 0.43$, or 43 percent of the decline in V-Dem scores can be explained by politically connected wealth inequality.

Table 4 Effects of decomposed wealth inequality and capital account openness on democracy

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: Overall sample							
Pol. Connected wealth inequality	0.0135	−0.00349**	−0.00381**	−0.00214*	−0.00324*	−0.00295**	−0.00643**
	(0.0753)	(0.00153)	(0.00159)	(0.00114)	(0.00193)	(0.00143)	(0.00310)
Pol. connected	−0.0215	0.00845***	0.00830***	0.00549**	0.00858**	0.00740***	0.0103
Wealth inequality*KAOPEN	(0.117)	(0.00299)	(0.00316)	(0.00244)	(0.00380)	(0.00227)	(0.00680)
Pol. Unconnected	−0.0545	−0.00330	−0.00414	−0.00249	−0.00792	−0.00262	−0.00950
Wealth Inequality	(0.0880)	(0.00518)	(0.00610)	(0.00337)	(0.00593)	(0.00409)	(0.00891)
Pol. Unconnected	0.0828	0.00361	0.00402	0.00240	0.00942	0.00272	0.0156
Wealth inequality*KAOPEN	(0.114)	(0.00643)	(0.00730)	(0.00424)	(0.00733)	(0.00506)	(0.0149)
N (No. of Countries)	758 (149)	758 (149)	758 (149)	758 (149)	758 (149)	758 (149)	758 (149)
R^2	0.096	0.110	0.097	0.150	0.118	0.141	0.153
Panel B: Democracies							
Pol. Connected wealth inequality	−0.0378	−0.00446**	−0.00506**	−0.00293*	−0.00474**	−0.00370*	−0.00681**
	(0.0962)	(0.00204)	(0.00196)	(0.00159)	(0.00232)	(0.00200)	(0.00327)
Pol. connected	0.0309	0.00974***	0.0100***	0.00668**	0.0107***	0.00852***	0.0115*
Wealth inequality*KAOPEN	(0.136)	(0.00331)	(0.00328)	(0.00267)	(0.00398)	(0.00276)	(0.00594)
Pol. unconnected	−0.0853	−0.00434	−0.00640	−0.00380	−0.01000	−0.00362	−0.00683
Wealth inequality	(0.102)	(0.00611)	(0.00740)	(0.00404)	(0.00710)	(0.00496)	(0.00755)
Pol. unconnected	0.0929	0.00370	0.00593	0.00314	0.0104	0.00322	0.00516
Wealth inequality*KAOPEN	(0.142)	(0.00775)	(0.00883)	(0.00532)	(0.00867)	(0.00633)	(0.0104)
N (No. of Countries)	542 (104)	542 (104)	542 (104)	542 (104)	542 (104)	542 (104)	542 (104)
R^2	0.082	0.133	0.131	0.183	0.138	0.173	0.159
Panel C: Countries with Billionaires							

Table 4 (continued)

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Pol. Connected Wealth Inequality	0.0371	−0.00163	−0.00269*	−0.000955	−0.00124	−0.00136	−0.00724**
	(0.0826)	(0.00151)	(0.00147)	(0.00107)	(0.00194)	(0.00124)	(0.00337)
Pol. connected	−0.0949	0.00624*	0.00675*	0.00407	0.00594	0.00568**	0.0108
Wealth inequality*KAOPEN	(0.142)	(0.00347)	(0.00353)	(0.00271)	(0.00437)	(0.00239)	(0.00737)
Pol. unconnected	−0.00563	−0.000652	−0.00272	−0.00100	−0.00550	−0.000651	−0.00949
Wealth inequality	(0.110)	(0.00587)	(0.00676)	(0.00392)	(0.00654)	(0.00453)	(0.00925)
Pol. unconnected	0.0571	0.000789	0.00257	0.000933	0.00687	0.0000791	0.0165
Wealth inequality*KAOPEN	(0.153)	(0.00796)	(0.00901)	(0.00525)	(0.00892)	(0.00601)	(0.0160)
N (No. of countries)	311 (56)	311 (56)	311 (56)	311 (56)	311 (56)	311 (56)	311 (56)
R^2	0.062	0.091	0.091	0.116	0.101	0.140	0.155
For all panels							
Country Controls and Country/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, capital account openness, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index. We use the first lag of all independent variables except for years of democracy. Politically unconnected wealth is the sum of wealth of all billionaires who are classified as politically unconnected

Table 5 Effects of wealth inequality and capital account openness on democracy in countries rated as democratic by polity

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: Wealth as percent of GDP							
Wealth Inequality	−0.0568	−0.00430*	−0.00549**	−0.00319*	−0.00671**	−0.00357*	−0.00670**
	(0.0568)	(0.00254)	(0.00275)	(0.00174)	(0.00294)	(0.00212)	(0.00329)
KAOPEN	−0.547	−0.0312	−0.0285	−0.0213	−0.0279	−0.0244	−0.0363
	(0.808)	(0.0306)	(0.0292)	(0.0212)	(0.0288)	(0.0202)	(0.0545)
W.Inequality*KAOPEN	0.0567	0.00678*	0.00789**	0.00485*	0.0100**	0.00592*	0.00836
	(0.0750)	(0.00378)	(0.00386)	(0.00277)	(0.00433)	(0.00309)	(0.00557)
R^2	0.082	0.130	0.127	0.179	0.133	0.167	0.158
Panel B: Inherited Wealth/GDP							
Inherited Wealth Inequality	−0.193	−0.00807	−0.0119	−0.00681	−0.0176	−0.00671	−0.0115
	(0.208)	(0.0131)	(0.0152)	(0.00818)	(0.0139)	(0.0104)	(0.0141)
KAOPEN	−0.732	−0.0305	−0.0277	−0.0209	−0.0288	−0.0231	−0.0337
	(0.833)	(0.0309)	(0.0295)	(0.0215)	(0.0290)	(0.0202)	(0.0552)
W.Inequality*KAOPEN	0.413	0.0147	0.0183	0.0111	0.0265	0.0119	0.0170
	(0.300)	(0.0162)	(0.0180)	(0.0109)	(0.0169)	(0.0131)	(0.0193)
R^2	0.085	0.129	0.126	0.178	0.134	0.165	0.157
Panel C: Pol. Connected Wealth/GDP							
Pol. Connected Wealth Inequality	−0.0491	−0.00503**	−0.00590***	−0.00343*	−0.00607**	−0.00417*	−0.00770**
	(0.0944)	(0.00229)	(0.00217)	(0.00183)	(0.00256)	(0.00229)	(0.00315)
KAOPEN	−0.496	−0.0274	−0.0236	−0.0184	−0.0204	−0.0211	−0.0317
	(0.787)	(0.0296)	(0.0284)	(0.0205)	(0.0280)	(0.0199)	(0.0530)
W.Inequality*KAOPEN	0.0446	0.0103***	0.0110***	0.00719**	0.0122***	0.00903***	0.0123**
	(0.131)	(0.00342)	(0.00335)	(0.00283)	(0.00403)	(0.00300)	(0.00588)
R^2	0.081	0.132	0.127	0.180	0.131	0.170	0.158

Table 5 (continued)

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
For all Panels							
N	542	542	542	542	542	542	542
No. of countries	104	104	104	104	104	104	104
Country/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index. We use the first lag of all independent variables except for years of democracy. Democracies are defined as any country which had a Polity score of 6 or higher in any of 1987, 1992, 1996, 2002, 2007, 2012, or 2017

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

this table clearly indicate that it is politically connected wealth inequality that matters for democracy.

This finding has important implications for other papers on inequality and democracy, such as Houle (2009), which does not break down inequality into its politically connected and unconnected components. Moreover, these results align with Bagchi and Svejnar (2015) which finds that politically connected wealth inequality lowers economic growth whereas politically unconnected wealth inequality does not, as well as Islam and McGillivray (2020) which finds that wealth inequality lowers growth only when governance is poor.

6 Robustness tests

Our baseline results suggest that although there is only weak evidence of a negative relationship between wealth inequality and democracy in the world at large, there is a negative relationship between politically connected wealth inequality and democracy, as measured by the various V-Dem indices and the continuous Machine Learning index. Having established these baseline results, we now turn to examining if those results are robust. We do so by imposing various sample restrictions, including different control variables, and considering alternative approaches of measuring wealth inequality. To preview the results, we find that the negative relationship between politically connected wealth inequality and democracy holds when we limit ourselves to countries that are classified as democratic, as well as when we control for taxes as a share of GDP or for economic growth. We also find that our conclusion regarding the lack of a robust relationship between wealth inequality and democracy holds when we use alternative measures of wealth inequality.

6.1 Various sample restrictions

6.1.1 Limiting ourselves to countries that are classified as democratic by polity

Polity defines a democracy as any country which scores 6 or higher on their measure. Therefore, we can test the effect of wealth inequality on already-existing democracies by restricting our sample only to countries that have been democratic at least once.[37] We do so in Table 5 where we find evidence that both overall and politically connected wealth inequality negatively impact the V-Dem scores of countries that are already democracies, but not their Polity scores. Here, our results indicate that a one standard deviation (3.9 percent of GDP) increase in wealth inequality is associated with a 0.0168 point decrease in Electoral Democracy in countries with no capital mobility, and a 0.01 point increase in Electoral Democracy when capital is fully mobile, although that latter effect is statistically indistinguishable from zero. Correspondingly, in column (2) of Panel C, we find that a one standard deviation increase in politically connected wealth inequality (2.29 percent of GDP) is associated with a 0.012 decline in Electoral Democracy scores in countries with

[37] We use Polity as our threshold for sample selection because although the translation from a continuous to a dichotomous measure of democracy is ad-hoc (Gründler & Krieger, 2022) we want to base our sample selection on a democracy measure which is not related to our inequality measure. We replicate these results using the dichotomous Machine Learning index in Online Appendix Table A.4.

no capital mobility indicating that the effect of overall inequality is being driven by politically connected inequality. Moreover, our results in Panels A and C suggest that wealth inequality and politically connected wealth inequality are particularly damaging for Liberal and Deliberative Democracy.

To give a concrete example, it is useful to look at the case of South Africa. From 2012 to 2017, South Africa's Electoral Democracy score declined from 0.78 to 0.735, while during the preceding five years their politically connected wealth inequality increased from 0 to 0.62 percent of GDP.[38] Given their Chinn-Ito index value of 0.17, we find that politically connected wealth inequality explains roughly 4.5 percent of the decline in Electoral Democracy from 2012 to 2017. While perhaps not the most important factor in explaining the erosion of democracy in South Africa, our results still indicate that politically connected wealth inequality plays a role in democratic reversals.

6.1.2 Limiting ourselves to countries that have billionaires at least once

Because our results may be driven by the large number of countries without billionaires, that is, the large number of countries for whom billionaire wealth is always 0, in Table 6 we run our regressions using only the 56 countries in our sample which have had a billionaire at least once. Our results partially confirm those in Table 3; we find no relationship between overall or inherited wealth inequality and democracy, but our negative relationship between politically connected wealth inequality and V-Dem scores is weaker. The Liberal Democracy index is the only V-Dem measure for which we find a significant, negative relationship between politically connected wealth inequality and democracy in this sample. To some extent, this result speaks to the decoupling of liberalism and democracy noted in Mounk (2018). That is, our results in this table indicate that recent declines in political freedom may represent an erosion in certain civil liberties rather than a decline in the core political rights of electoral democracy. Additionally, we also confirm the negative and statistically significant relationship between politically connected wealth inequality and the continuous Machine Learning index of democracy in this restricted sample, with the estimated coefficient similar in magnitude across Tables 3 and 6.

6.2 Inclusion of additional controls

6.2.1 Including taxes as a percentage of GDP as an additional control

We recognize that we have not included all theoretical determinants of democracy proposed in the literature. While time-invariant factors cannot be included in the presence of country fixed effects, we are able to include a control for taxes as a percentage of GDP, data for which are provided by the World Bank. The inclusion of this control variable is motivated by the work of Ross (2001) who notes that a state that is rich in natural resources could in theory use low tax rates and high spending to dampen pressures for democracy. Thus, to the extent that high rents from natural resources accompany low levels of taxes, and thwart democratization by making the rulers less responsive to the preferences of citizens, we ought to include taxes as a percentage of GDP as a control. We do so in Table 7 and although we lose 30 countries and over 300

[38] Over that same time period, overall billionaire wealth in South Africa increased from 2.79 to 4.07 percent of GDP.

Table 6 Effects of wealth inequality and capital account openness on democracy in countries with at least one billionaire

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: Wealth as percent of GDP							
Wealth Inequality	0.0218	−0.00129	−0.00273	−0.000990	−0.00284	−0.00112	−0.00806**
	(0.0627)	(0.00250)	(0.00278)	(0.00167)	(0.00276)	(0.00194)	(0.00372)
KAOPEN	0.322	−0.00550	−0.0130	−0.000214	−0.00495	0.00102	−0.0448
	(1.382)	(0.0472)	(0.0460)	(0.0352)	(0.0440)	(0.0306)	(0.0787)
W.Inequality*KAOPEN	−0.0167	0.00331	0.00432	0.00224	0.00548	0.00260	0.0136*
	(0.0812)	(0.00390)	(0.00418)	(0.00269)	(0.00428)	(0.00301)	(0.00727)
R^2	0.057	0.086	0.086	0.110	0.096	0.128	0.154
Panel B: Inherited Wealth/GDP							
Inherited Wealth Inequality	−0.0244	−0.00122	−0.00496	−0.00127	−0.00923	−0.00114	−0.0157
	(0.181)	(0.0116)	(0.0135)	(0.00737)	(0.0125)	(0.00900)	(0.0154)
KAOPEN	−0.0232	−0.00168	−0.00884	0.00258	−0.00697	0.00565	−0.0433
	(1.413)	(0.0474)	(0.0471)	(0.0352)	(0.0454)	(0.0304)	(0.0809)
W.Inequality*KAOPEN	0.166	0.00414	0.00730	0.00288	0.0143	0.00246	0.0292
	(0.255)	(0.0147)	(0.0169)	(0.00945)	(0.0159)	(0.0113)	(0.0248)
R^2	0.067	0.084	0.084	0.108	0.097	0.124	0.156
Panel C: Pol. Connected Wealth/GDP							
Pol. Connected Wealth Inequality	0.0406	−0.00168	−0.00297*	−0.00106	−0.00167	−0.00147	−0.00762**
	(0.0836)	(0.00160)	(0.00160)	(0.00114)	(0.00187)	(0.00132)	(0.00328)
KAOPEN	0.379	−0.00193	−0.00745	0.00236	0.00547	0.00299	−0.0166
	(1.262)	(0.0427)	(0.0414)	(0.0319)	(0.0397)	(0.0284)	(0.0751)
W.Inequality*KAOPEN	−0.0948	0.00631*	0.00707**	0.00419	0.00657	0.00576**	0.0119*
	(0.137)	(0.00337)	(0.00342)	(0.00265)	(0.00408)	(0.00236)	(0.00667)
R^2	0.059	0.091	0.090	0.115	0.097	0.139	0.142

Table 6 (continued)

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
For all panels							
N	311	311	311	311	311	311	311
No. of countries	56	56	56	56	56	56	56
Country/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index. We use the first lag of all independent variables except for years of democracy. We include countries that have had a billionaire at least once. If a country has a billionaire in some years but not others, we include that country with a 0 for billionaire wealth rather than dropping it from the regression

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

observations, our results are largely unchanged. When looking at overall inequality in Panel A, we find a statistically significant, negative relationship between overall wealth inequality and some of the V-Dem indices, but not Polity scores. Our results for politically connected wealth inequality in Panel C continue to be negative and statistically significant, and now we also find evidence of a negative relationship between politically connected wealth inequality and Polity scores in col. (1).

6.2.2 Controlling for the growth rate of the economy

Because of the way we measure wealth inequality, the sum of billionaire wealth normalized by GDP, economic downturns may decrease our measure by bringing about even a small decline in billionaire wealth below the $1 billion threshold needed for inclusion in the *Forbes* list. The relationship between inequality and democracy that we find may therefore be biased by the fact that an economic downturn could both reduce inequality and, simultaneously, trigger regime change. Although the relationship between economic growth and the growth of political rights and civil liberties is highly dependent on prior regime type (Wintrobe, 1990; Islam & Winer, 2004), and hence may not bias our results strictly upwards or downwards, including the growth rate of real GDP per capita can be useful in addressing the confounding impact of economic growth on both inequality and democracy. Because the relationship between democratization and growth is highly endogenous (Acemoglu et al., 2019), we lag growth rates. Data for the growth of real GDP per capita comes from the Penn World Tables. Our results in Table 8 are marginally stronger when controlling for growth, as the relationship between politically connected wealth inequality and participatory democracy is now significant at the 5 percent level in column (4) of Panel C, rather than at the 10 percent level, which it is in column (4) of Panel C in Table 3. Our coefficient magnitudes are also slightly larger in columns (2)–(7) in Panel C although a formal t-test indicates that they are statistically indistinguishable from the comparable coefficients in Table 3. Most importantly though, the results from this table assure us that our results are not being driven by economic downturns leading to billionaires dropping from the *Forbes* list. Instead it is being driven by the underlying relationship between politically connected wealth inequality and democracy.

6.3 Alternative wealth inequality measures

We recognize that billionaire wealth as a percentage of GDP is a relatively unconventional measure of wealth inequality, with its first use in Bagchi and Svejnar (2015). To test if our generally lackluster results for overall wealth inequality could be attributed to our unusual measure, we also examine the effect of the top percentile and decile wealth shares from the World Inequality Database (WID) on measures of democracy. For the vast majority of countries, the use of data on wealth inequality from the WID requires us to drop observations for 1987 and 1992, but there are a few countries, such as the U.S. and France, for which the 1987 and 1992 observations are available. This dataset gives us 567 observations across 147 countries. As can be seen in Table 1, the top percentile and decile wealth shares have means of 0.305 and 0.638, respectively, indicating that the top percentile and top decile hold about 31 percent and 64 percent of a country's wealth on average.

Table 7 Effects of Wealth Inequality and Capital Account Openness on Democracy, Controlling for Taxes as a Share of GDP

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: Wealth as percent of GDP							
Wealth inequality	−0.0887	−0.00562	−0.00491	−0.00406*	−0.00565*	−0.00411	−0.00817**
	(0.0595)	(0.00339)	(0.00297)	(0.00223)	(0.00310)	(0.00271)	(0.00396)
KAOPEN	−0.700	−0.0618*	−0.0555	−0.0383	−0.0638*	−0.0500**	−0.0448
	(1.018)	(0.0370)	(0.0353)	(0.0270)	(0.0361)	(0.0250)	(0.0658)
W.Inequality*KAOPEN	0.106	0.00842**	0.00718*	0.00603**	0.00937**	0.00654*	0.0142*
	(0.0689)	(0.00421)	(0.00382)	(0.00284)	(0.00401)	(0.00335)	(0.00740)
R^2	0.117	0.155	0.155	0.188	0.164	0.199	0.154
Panel B: Inherited wealth/GDP							
Inherited Wealth Inequality	0.00189	−0.0000102	0.00253	−0.00126	−0.00387	0.00112	−0.0165
	(0.140)	(0.0131)	(0.0116)	(0.00852)	(0.0127)	(0.0112)	(0.0140)
KAOPEN	−0.644	−0.0506	−0.0438	−0.0311	−0.0557	−0.0401	−0.0464
	(1.036)	(0.0372)	(0.0351)	(0.0272)	(0.0363)	(0.0248)	(0.0671)
W.Inequality*KAOPEN	0.102	0.00290	−0.000725	0.00340	0.00939	0.000776	0.0320
	(0.160)	(0.0146)	(0.0132)	(0.00974)	(0.0143)	(0.0127)	(0.0237)
R^2	0.118	0.147	0.148	0.179	0.156	0.189	0.157
Panel C: Pol. Connected wealth/GDP							
Pol. Connected wealth inequality	−0.162***	−0.0120***	−0.0115***	−0.00816***	−0.00950***	−0.00948***	−0.00877**
	(0.0595)	(0.00245)	(0.00209)	(0.00176)	(0.00280)	(0.00212)	(0.00364)
KAOPEN	−0.669	−0.0618*	−0.0570*	−0.0377	−0.0607*	−0.0507**	−0.0306
	(0.989)	(0.0352)	(0.0338)	(0.0256)	(0.0345)	(0.0242)	(0.0646)
W.Inequality*KAOPEN	0.200**	0.0196***	0.0186***	0.0132***	0.0178***	0.0162***	0.0149***
	(0.0774)	(0.00312)	(0.00271)	(0.00229)	(0.00349)	(0.00260)	(0.00535)
R^2	0.118	0.165	0.167	0.196	0.170	0.216	0.147

Table 7 (continued)

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
For all Panels							
N	447	447	447	447	447	447	447
No. of countries	119	119	119	119	119	119	119
Country/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, taxes as a percentage of GDP, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index. We use the first lag of all independent variables except for years of democracy

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Looking at Table 9, we see that there is an insignificant relationship between top percentile wealth shares and all seven measures of democracy, regardless of whether we exclude capital account openness (Panel A) or we include it, interacted with wealth inequality (Panel C). Likewise, we fail to find any statistically significant relationship between top decile wealth shares and all measures of democracy, regardless of whether we exclude capital account openness (Panel B) or we include it, interacted with wealth inequality (Panel D). These results indicate that only looking at total wealth inequality misses the importance of how that wealth was acquired. Because the WID wealth inequality measures do not—*and cannot*—distinguish between politically connected and unconnected wealth, it is unable to tease out the importance of different types of wealth, which we are able to do using our novel measures. Furthermore, the null result using data from the WID is not simply a function of the lack of observations on wealth shares for 1987 and 1992. In Online Appendix Table A.3, when we estimate our original specification using only those observations for which we have wealth share data from the WID, we find that while there is no relationship between overall inequality and V-Dem scores, there continues to be a negative and significant relationship between politically connected inequality and V-Dem scores (for four of the five indices) as well as the continuous Machine Learning index.[39]

Beyond looking at alternative data sources, we also recognize that the use of a \$1 billion threshold for inclusion on the *Forbes* list is arbitrary. In principle, individuals could move in and out of the *Forbes* list because of minor perturbations in wealth around the \$1 billion threshold. To address this concern, we increase the threshold needed for inclusion on the list of billionaires to \$1.5 billion in Panel A, \$2 billion in Panel B, \$2.5 billion in Panel C, and \$3 billion in Panel D of Table 10. In all four panels, we find that there is a negative and statistically significant relationship between politically connected wealth inequality and V-Dem's electoral democracy index, liberal democracy index, and deliberative democracy index, as well as the continuous Machine Learning index. Although we lose statistical significance in Panels A and B for the participatory and egalitarian democracy index, we regain statistical significance in Panels C and D when considering either a \$2.5 billion threshold or a \$3 billion threshold for inclusion. In the Online Appendix, we show that we obtain statistically significant results for politically connected wealth inequality when changing the threshold for inclusion in finer increments of \$250 million, and additionally confirm the null results for overall and inherited wealth inequality.[40] Thus, we are confident that our results are not driven by *Forbes'* decision to use \$1 billion as the threshold for being included on the billionaire list.

[39] In additional checks not included in the paper, we confirm that our baseline results in Table 3 survive the introduction of each measure of wealth inequality or income inequality from the WID, such as the top percentile wealth share, the top decile wealth share, the Gini coefficient of wealth, etc. as controls.

[40] See Online Appendix Tables A.9 and A.10 for overall, A.11 and A.12 for inherited, and A.13 and A.14 for politically connected wealth inequality results. Table A.15 shows that the results for politically connected wealth inequality hold for democracies as well.

Table 8 Effects of wealth inequality and capital account openness on democracy, controlling for real GDP per capita growth

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: Wealth as percent of GDP							
Wealth Inequality	−0.0156	−0.00331	−0.00384*	−0.00224	−0.00499**	−0.00271	−0.00774**
	(0.0475)	(0.00203)	(0.00216)	(0.00138)	(0.00232)	(0.00167)	(0.00349)
KAOPEN	−0.254	−0.0243	−0.0221	−0.0147	−0.0302	−0.0216	−0.0133
	(0.762)	(0.0263)	(0.0248)	(0.0186)	(0.0244)	(0.0171)	(0.0494)
W.Inequality*KAOPEN	0.0309	0.00546*	0.00556*	0.00357*	0.00797**	0.00453*	0.0130*
	(0.0626)	(0.00298)	(0.00310)	(0.00214)	(0.00339)	(0.00243)	(0.00695)
R^2	0.097	0.108	0.094	0.149	0.115	0.137	0.153
Panel B: Inherited wealth/GDP							
Inherited Wealth Inequality	−0.0783	−0.00551	−0.00720	−0.00387	−0.0128	−0.00430	−0.0160
	(0.182)	(0.0107)	(0.0124)	(0.00674)	(0.0116)	(0.00834)	(0.0147)
KAOPEN	−0.365	−0.0224	−0.0199	−0.0134	−0.0301	−0.0194	−0.0132
	(0.764)	(0.0262)	(0.0249)	(0.0186)	(0.0244)	(0.0169)	(0.0501)
W.Inequality*KAOPEN	0.218	0.00942	0.0101	0.00629	0.0191	0.00691	0.0289
	(0.236)	(0.0127)	(0.0145)	(0.00826)	(0.0138)	(0.0100)	(0.0235)
R^2	0.099	0.107	0.092	0.147	0.115	0.133	0.153
Panel C: Pol. Connected wealth/GDP							
Pol. connected wealth inequality	0.00313	−0.00394**	−0.00440***	−0.00257**	−0.00437**	−0.00330**	−0.00769***
	(0.0751)	(0.00166)	(0.00161)	(0.00129)	(0.00189)	(0.00160)	(0.00271)
KAOPEN	−0.206	−0.0214	−0.0190	−0.0127	−0.0241	−0.0193	−0.00198
	(0.744)	(0.0255)	(0.0241)	(0.0181)	(0.0237)	(0.0168)	(0.0486)
W.Inequality*KAOPEN	−0.00709	0.00902***	0.00901***	0.00599**	0.0101***	0.00782***	0.0123**
	(0.114)	(0.00293)	(0.00304)	(0.00250)	(0.00352)	(0.00234)	(0.00596)
R^2	0.097	0.110	0.096	0.150	0.114	0.140	0.149

Table 8 (continued)

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
For all panels							
N	758	758	758	758	758	758	758
No. of Countries	149	149	149	149	149	149	149
Country/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, growth of real GDP per capita, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index. We use the first lag of all independent variables except for years of democracy

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9 Effects of Top wealth shares and capital account openness on democracy

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: Top Percentile wealth shares no interaction							
Top Percentile Wealth Share	0.976	−0.0430	−0.0879	−0.0363	−0.0834	−0.0446	0.0206
	(3.043)	(0.104)	(0.0974)	(0.0747)	(0.117)	(0.0679)	(0.281)
R^2	0.040	0.023	0.023	0.024	0.049	0.034	0.047
Panel B: Top decile wealth shares							
Top Decile Wealth Share	0.803	−0.0613	−0.0967	−0.0407	−0.106	−0.0496	0.0304
	(3.060)	(0.108)	(0.103)	(0.0766)	(0.126)	(0.0707)	(0.272)
R^2	0.040	0.023	0.023	0.024	0.049	0.034	0.047
Panel C: Top percentile wealth shares							
Top Percentile Wealth Share	0.718	−0.0598	−0.0828	−0.0209	−0.113	−0.0746	−0.170
	(3.521)	(0.125)	(0.121)	(0.0875)	(0.140)	(0.0833)	(0.340)
KAOPEN	−0.514	−0.0308	−0.0118	0.00594	−0.0384	−0.0291	−0.136
	(1.110)	(0.0429)	(0.0442)	(0.0319)	(0.0522)	(0.0334)	(0.0915)
Top Percentile*KAOPEN	0.574	0.0374	−0.0113	−0.0344	0.0662	0.0669	0.425
	(3.915)	(0.133)	(0.138)	(0.0966)	(0.162)	(0.0981)	(0.321)
R^2	0.040	0.023	0.023	0.024	0.049	0.035	0.050
Panel D: Top decile wealth shares							
Top decile wealth share	0.592	−0.0695	−0.0820	−0.0207	−0.115	−0.0744	−0.130
	(3.578)	(0.131)	(0.127)	(0.0910)	(0.150)	(0.0858)	(0.333)
KAOPEN	−0.632	−0.0309	0.00544	0.0238	−0.0319	−0.0440	−0.234
	(2.378)	(0.0838)	(0.0867)	(0.0599)	(0.104)	(0.0626)	(0.192)
Top Decile*KAOPEN	0.472	0.0183	−0.0329	−0.0447	0.0214	0.0553	0.358
	(3.978)	(0.132)	(0.137)	(0.0935)	(0.163)	(0.0959)	(0.313)
R^2	0.040	0.023	0.023	0.025	0.049	0.035	0.049
For all Panels							
N	567	567	567	567	567	567	567
No. of countries	147	147	147	147	147	147	147
Country/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index. We use the first lag of all independent variables except for years of democracy

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10 Effects of Politically connected wealth inequality on democracy at varying thresholds

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
Panel A: $1.5 Billion threshold							
Pol. connected wealth inequality	0.0389	−0.00320*	−0.00348**	−0.00180	−0.00320*	−0.00260	−0.00782***
Threshold: $1.5 billion	(0.0780)	(0.00183)	(0.00170)	(0.00133)	(0.00181)	(0.00163)	(0.00257)
W.Inequality*KAOPEN	−0.0496	0.00811**	0.00796**	0.00501*	0.00884**	0.00690***	0.0127**
	(0.121)	(0.00319)	(0.00320)	(0.00268)	(0.00345)	(0.00246)	(0.00581)
R^2	0.096	0.109	0.094	0.148	0.113	0.138	0.149
Panel B: $2 Billion threshold							
Pol. Connected wealth inequality	0.0297	−0.00369*	−0.00417**	−0.00208	−0.00406**	−0.00291	−0.00886***
Threshold: $2 billion	(0.0882)	(0.00209)	(0.00194)	(0.00161)	(0.00194)	(0.00196)	(0.00286)
W.Inequality*KAOPEN	−0.0285	0.00906***	0.00917***	0.00567**	0.0105***	0.00755***	0.0149***
	(0.122)	(0.00302)	(0.00298)	(0.00254)	(0.00298)	(0.00258)	(0.00507)
R^2	0.096	0.110	0.095	0.149	0.114	0.139	0.150
Panel C: $2.5 Billion threshold							
Pol. connected wealth Inequality	−0.0412	−0.00687***	−0.00674***	−0.00397**	−0.00636*	−0.00539**	−0.0128***
Threshold: $2.5 billion	(0.0949)	(0.00233)	(0.00202)	(0.00185)	(0.00344)	(0.00220)	(0.00421)
W.Inequality*KAOPEN	0.0700	0.0132***	0.0128***	0.00848***	0.0134***	0.0105***	0.0192***
	(0.121)	(0.00309)	(0.00274)	(0.00245)	(0.00453)	(0.00284)	(0.00650)
R^2	0.096	0.112	0.098	0.152	0.115	0.141	0.150
Panel D: $3 Billion threshold							
Pol. connected wealth inequality	−0.0578	−0.00782***	−0.00739***	−0.00444**	−0.00790*	−0.00622**	−0.0147***
Threshold: $3 billion	(0.102)	(0.00271)	(0.00236)	(0.00220)	(0.00413)	(0.00261)	(0.00448)
W.Inequality*KAOPEN	0.0906	0.0144***	0.0137***	0.00915***	0.0155***	0.0116***	0.0217***
	(0.130)	(0.00353)	(0.00310)	(0.00284)	(0.00529)	(0.00332)	(0.00674)
R^2	0.096	0.112	0.098	0.152	0.117	0.142	0.151

Table 10 (continued)

	(1) Polity Year	(2) Electoral Democracy	(3) Liberal Democracy	(4) Participatory Democracy	(5) Deliberative Democracy	(6) Egalitarian Democracy	(7) Machine Learning
For all panels							
N	758	758	758	758	758	758	758
No. of countries	149	149	149	149	149	149	149
Country/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors are clustered at the country level. All estimations include country and year fixed effects and include GDP per capita, the UN Education Index, years of democracy, oil rents, capital account openness, and the average Freedom House scores of neighboring countries as controls. Column 1 uses democracy scores from Polity. Columns 2 through 6 use democracy measures from the V-Dem dataset. Column 7 uses the continuous Machine Learning index. We use the first lag of all independent variables except for years of democracy. Each threshold determines the wealth a billionaire must have to be counted in our measure of wealth inequality. Hence, a threshold of $1.5 billion means that we exclude all billionaires whose wealth is less than $1.5 billion. Correspondingly, our regressions in Tables 3, 4, 5, 6, 7, and 8 use a threshold of $1 billion

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11 Countries used

Countries with at least one billionaire						
Argentina	Australia	Austria	Bahrain	Belgium	Brazil	Canada
Chile	China	Colombia	Cyprus	Czechia	Denmark	Ecuador
Egypt	Finland	France	Georgia	Germany	Greece	India
Indonesia	Ireland	Israel	Italy	Japan	Kazakhstan	Kuwait
Lebanon	Malaysia	Mexico	Morocco	Netherlands	New Zealand	Nigeria
Norway	Peru	Philippines	Poland	Portugal	Romania	Russia
Saudi Arabia	Singapore	South Africa	South Korea	Spain	Sweden	Switzerland
Thailand	Turkey	Ukraine	UAE	United Kingdom	United States	Venezuela
Countries rated "Democratic" at least once						
Albania	Argentina	Armenia	Australia	Austria	Bangladesh	Belarus
Belgium	Benin	Bolivia	Botswana	Brazil	Bulgaria	Burkina Faso
Burundi	Canada	Cape Verde	Central African Republic	Chile	Colombia	Comoros
Costa Rica	Croatia	Cyprus	Czechia	Denmark	Dominican Republic	Ecuador
El Salvador	Estonia	Finland	France	Gambia	Georgia	Germany
Ghana	Greece	Guatemala	Guinea-Bissau	Haiti	Honduras	Hungary
India	Indonesia	Iraq	Ireland	Israel	Italy	Jamaica
Japan	Kyrgyzstan	Latvia	Lebanon	Lesotho	Liberia	Lithuania
Macedonia	Madagascar	Malawi	Malaysia	Mali	Mauritius	Mexico
Moldova	Mongolia	Myanmar	Namibia	Nepal	Netherlands	New Zealand
Nicaragua	Niger	Nigeria	Norway	Pakistan	Panama	Paraguay
Peru	Philippines	Poland	Portugal	Romania	Russia	Senegal
Sierra Leone	Slovak Republic	Slovenia	South Africa	South Korea	Spain	Sri Lanka
Sudan	Sweden	Switzerland	Thailand	Trinidad and Tobago	Tunisia	Turkey
Ukraine	United Kingdom	United States	Uruguay	Venezuela	Zambia	

7 Conclusion

The literature examining the relationship between inequality and democracy is large and is yet to arrive at a consensus (Rød et al., 2020; Ahlquist & Wibbels, 2012). The conflicting findings reported in the literature likely stem from the use of various different proxies of inequality that capture wealth inequality very imperfectly, the weak link between the available data and the theoretical arguments, and the difficulty of identifying the causal mechanisms linking inequality and democracy. We view our paper as advancing this literature in three ways. First, in contrast to much of the literature that uses proxies of income inequality, we exploit a novel measure of wealth inequality developed first in Bagchi and Svejnar (2015) and expanded subsequently in Bagchi et al. (2019) to re-examine the relationship between inequality and democracy. Second, we emphasize the rich heterogeneity within economic elites and dig deeper to understand what gives rise to inequality in the first place. Third, we tie our empirical results to a key theoretical channel proposed in the literature—capital mobility—that is likely to moderate the relationship between inequality and democracy.

Our results reveal that inherited and politically unconnected wealth inequality are not important determinants of democracy, as evinced by the lack of any relationship between those measures of wealth inequality and the different measures of democracy we examine. While there is some evidence of a negative relationship between wealth inequality and democracy, that relationship appears to be driven by politically connected wealth inequality, which has a robust negative impact on democracy when measured by various V-Dem indices or the continuous Machine Learning index (Gründler & Krieger, 2021). Our interpretation of these results is that billionaires who inherit their wealth or those who owe their fortunes to success in a market economy are significantly different from other types of economic elites who may co-opt the political system to entrench themselves. Thus, while billionaires who owe their wealth to their entrepreneurial foresight or to the foresight of past generations do not pose a threat to democracy as best as we can say, billionaires who leveraged political connections for economic gain are likely to have a negative influence on democracy. The heterogeneity across billionaires may well be the most important take-away from our paper in that it suggests that when scholars examine the relationship between inequality and democracy they should pay attention to what gave rise to inequality in the first place. Our findings strongly suggest that billionaires are heterogeneous, and to lump them together as a class is to miss the rich variation in the sources of their wealth. The variation in the source of billionaire wealth has been shown to be important in quantifying the effects of wealth inequality on economic growth (Bagchi & Svejnar, 2015), and now our paper establishes that those differences in the source of billionaire wealth also influence the impact of wealth inequality on democracy. We find it likely that the re-emergence of politically connected wealth in many countries, including those that would be classified as democratic, may be associated with the ongoing wave of autocratization, a phenomenon noted in Lührmann and Lindberg (2019).

In terms of underlying mechanisms that explain the differences in the impact of various types of wealth inequality, we theorize that politically connected economic elites are different in that they may have much to fear from a dismantling of the existing order. Given that their rise to affluence may have been inextricably linked with the existing political regime, it would be rational for them to prop up the political incumbents who would prevent an opening up of the economy. In contrast, for a newly emerging class of economic elites—perhaps those belonging to sectors where government intervention is relatively muted—democracy might be the preferred alternative. Although there is the risk of the median voter voting for tax rates that are too high, as proposed by Meltzer and Richard (1981), these elites might still find that prospect more palatable because a democratic set-up involving the rule of law with checks and balances and a healthy respect for property rights is likely to better protect their wealth from expropriation.

Additionally, an angle that we do not explore in this paper but could be an area for future research is that high levels of politically connected inequality may lead to a loss of faith in democracy as a means of providing economic mobility and avenues for social advancement—a possibility hinted at in Foa and Mounk (2016) who note that "Citizens in a number of supposedly consolidated democracies in North America and Western Europe....have also become more cynical about the value of democracy as a political system, less hopeful that anything they do might influence public policy, and more willing to express support for authoritarian alternatives." We speculate that such cynicism may be particularly high in countries where political connections are viewed as indispensable to success in the economic arena and this erosion of favorable attitudes towards democracy might ultimately drive part of the negative relationship between politically connected inequality and democracy. Along related lines, new research is beginning to find that social mobility, not just inequality, plays a role in civil unrest (Houle, 2019). These findings suggest that perhaps a key way of fighting against democratic breakdown is not by reducing the wealth of market actors, but by putting measures in place that would thwart the use of political power to enrich certain businesses at the expense of others and instituting a principle of generality in politics as Buchanan and Congleton (1998) propose. Thus the answer to the puzzle of democratic breakdown is not likely to be achieved by the introduction of new taxes on wealth or selective enforcement of our antitrust laws but may instead hinge on preventing quid pro quo arrangements between market participants and political actors.

Going forward, future researchers could benefit from a disaggregated approach to measuring wealth inequality. One could break down the lobbying by billionaires into lobbying from those who are self-made, inherited, or politically connected to better assess the effects of different types of lobbying and their subsequent impact on policy. Billionaires, and more generally economic elites, can influence policy in democracies without causing an explicit breakdown of democratic rights and norms, and future researchers may want to explore how wealth inequality impinges on economic freedom. We view our work as taking some preliminary steps in uncovering these issues but we raise as many questions as answers and clearly, much more remains to be done.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11127-023-01082-9.

Acknowledgements We thank Peter Calcagno, Joshua Hall, Christopher Kilby, Laura Meinzen-Dick, Olukunle Owolabi, Daniel Treisman, two anonymous referees, and participants at a session of the 2019 National Tax Association Conference for their helpful comments.

References

Acemoglu, D., Naidu, S., Restrepo, P., & Robinson, J. A. (2019). Democracy does cause growth. *Journal of Political Economy, 127*(1), 47–100.

Acemoglu, D., & Robinson, J. A. (2006). *Economic origins of dictatorship and democracy*. New York: Cambridge University Press.

Ahlquist, J. S., & Wibbels, E. (2012). Riding the wave: World trade and factor-based models of democratization. *American Journal of Political Science, 56*(2), 447–464.

Albertus, M., & Menaldo, V. (2018). *Authoritarianism and the elite origins of democracy*. Cambridge: Cambridge University Press.
Ansell, B., & Samuels, D. (2010). Inequality and democratization: A contractarian approach. *Comparative Political Studies, 43*(12), 1543–1574.
Bagchi, S., Curran, M., & Fagerstrom, M. J. (2019). Monetary growth and wealth inequality. *Economics Letters, 182*, 23–25.
Bagchi, S., & Svejnar, J. (2015). Does wealth inequality matter for growth? The effect of billionaire wealth, income distribution, and poverty. *Journal of Comparative Economics, 43*(3), 505–530.
Bagchi, S. & Svejnar, J. (2016). *Inequality and growth: Patterns and policy*, chapter Does Wealth Distribution and the Source of Wealth Matter for Economic Growth? Inherited v. Uninherited Billionaire Wealth and Billionaires' Political Connections, pp. 163–194. Palgrave Macmillan, UK.
Baltagi, B. H., Demetriades, P. O., & Law, S. H. (2009). Financial development and openness: Evidence from panel data. *Journal of Development Economics, 89*(2), 285–296.
Bernhard, M., Reenock, C., & Nordstrom, T. (2004). The legacy of western overseas colonialism on democratic survival. *International Studies Quarterly, 48*(1), 225–250.
Black, B. S., Kraakman, R., & Tarassova, A. (2000). Russian privatization and corporate governance: What went wrong? *Stanford Law Review, 52*(6), 1731–1808.
Boix, C. (2003). *Democracy and redistribution*. New York: Cambridge University Press.
Buchanan, J. M., & Congleton, R. D. (1998). *Politics by principle, not interest: Towards nondiscriminatory democracy*. New York: Cambridge University Press.
Buchanan, J. M. & Tullock, G. (1999). *The calculus of consent*. Liberty Fund.
Cheibub, J. A., Gandhi, J., & Vreeland, J. R. (2010). Democracy and dictatorship revisited. *Public Choice, 143*, 67–101.
Chinn, M. D., & Ito, H. (2006). What matters for financial development? Capital controls, institutions, and interactions. *Journal of Development Economics, 81*(1), 163–192.
Coppedge, M., Edgell, A. B., Knutsen, C. H., & Lindberg, S. I. (2022a). *Why democracies develop and decline*, chapter V-dem reconsiders democratization, pp. 1–28. Cambridge University Press.
Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., Altman, D., Bernhard, M., Cornell, A., Fish, M. S., Gastaldi, L., Gjerløw, H., Glynn, A., Grahn, S., Hicken, A., Kinzelbach, K., Marquardt, K. L., McMann, K., Mechkova, V., Paxton, P., Pemstein, D., von Römer, J., Seim, B., Sigman, R., Skaaning, S.-E., Staton, J., Tzelgov, E., Uberti, L., ting Wang, Y., Wig, T., & Ziblatt, D. (2022b). V-Dem Codebook v12. *Varieties of democracy (V-Dem) project*.
Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., Marquardt, K. L., Medzihorsky, J., Pemstein, D., Alizada, N., Gastaldi, L., Hindle, G., Pernes, J., von Römer, J., Tzelgov, E., ting Wang, Y., & Wilson, S. (2022c). V-dem methodology v12. *Varieties of democracy (V-Dem) project*.
Dahl, R. A. (1972). *Polyarchy: Participation and opposition*. Yale University Press.
Dash, B. B., Ferris, J. S., & Voia, M.-C. (2023). Inequality, transaction costs and voter turnout: Evidence from Canadian provinces and Indian states. *Public Choice, 194*, 325–346.
Diamond, L. (2015). Facing up to the democratic recession. *Journal of Democracy, 26*(1), 141–155.
Easterly, W. (2007). Inequality does cause underdevelopment: Insights from a new instrument. *Journal of Development Economics, 84*(2), 755–776.
Ekelund, R. B., & Thornton, M. (2020). Rent seeking as an evolving process: The case of the Ancien Régime. *Public Choice, 182*, 139–155.
Faccio, M. (2006). Politically connected firms. *American Economic Review, 96*(1), 369–386.
Fisman, R. (2001). Estimating the value of political connections. *American Economic Review, 91*(4), 1095–1102.
Foa, R. S., & Mounk, Y. (2016). The danger of deconsolidation: The democratic disconnect. *Journal of Democracy, 27*(3), 5–17.
Gorodnichenko, Y., & Roland, G. (2021). Culture, institutions and democratization. *Public Choice, 187*, 165–195.
Gründler, K., & Krieger, T. (2021). Using machine learning for measuring democracy: A practicioners guide and a new updated dataset for 186 Countries from 1919 to 2019. *European Journal of Political Economy*, 70.
Gründler, K. & Krieger, T. (2022). Should we care (more) about data aggregation? *European Economic Review*, 142.
Holcombe, R. G. (2018). *Political capitalism: How economic and political power is made and maintained*. Cambridge University Press.
Houle, C. (2009). Inequality and democracy: Why inequality harms consolidation but does not affect democratization. *World Politics, 61*(4), 589–622.

Houle, C. (2016). Inequality, economic development, and democratization. *Studies in Comparative International Development, 51*(4), 503–529.

Houle, C. (2019). Social mobility and political instability. *Journal of Conflict Resolution, 63*(1), 85–111.

Husted, T. A., & Kenny, L. W. (1997). The effect of the expansion of the voting franchise on the size of government. *Journal of Political Economy, 105*(1), 54–82.

Högström, J. (2013). Does the choice of democracy measure matter? Comparisons between the two leading democracy indices, Freedom House and Polity IV. *Government and Opposition, 48*(2), 201–221.

Ilzetzki, E., Reinhart, C. M., & Rogoff, K. S. (2019). Exchange arrangements entering the twenty-first century: Which anchor will hold? *The Quarterly Journal of Economics, 134*(2), 599–646.

Islam, M. N., & Winer, S. L. (2004). Tinpots, totalitarians (and democrats): An empirical investigation of the effects of economic growth on civil liberties and political rights. *Public Choice, 118*(3/4), 289–323.

Islam, M. R., & McGillivray, M. (2020). Wealth inequality, governance and economic growth. *Economic Modelling, 88*, 1–13.

Kaufman, R. R. (2009). The political effects of inequality in Latin America: Some inconvenient facts. *Comparative Politics, 41*(3), 359–379.

Knutsen, C. H. & Dahlum, S. (2022). *Why democracies develop and decline*, chapter Economic Determinants, pp. 119–160. Cambridge University Press.

Krieckhaus, J., Byunghwan, S., Bellinger, N. M., & Wells, J. M. (2014). Economic inequality and democratic support. *The Journal of Politics, 76*(1), 139–151.

Krieger, T., & Meierrieks, D. (2016). Political capitalism: The interaction between income inequality, economic freedom and democracy. *European Journal of Political Economy, 45*, 115–132.

Lambsdorff, J. G. (2002). Corruption and rent-seeking. *Public Choice, 113*, 97–125.

Leeson, P. T. (2005). Endogenizing fractionalization. *Journal of Institutional Economics, 1*(1), 75–98.

Leeson, P. T., & Dean, A. M. (2009). The democratic domino theory: An empirical investigation. *American Journal of Political Science, 53*(3), 533–551.

Lührmann, A., & Lindberg, S. I. (2019). A third wave of autocratization is here: What is new about it? *Democratization, 26*(7), 1095–1113.

Marshall, M. G., Gurr, T. R., & Jaggers, K. (2017). *Polity IV project dataset users' manual*. Center for Systemic Peace.

Meltzer, A. H., & Richard, S. F. (1981). A rational theory of the size of government. *Journal of Political Economy, 89*(5), 914–927.

Midlarsky, M. I. (1992). The origins of democracy in agrarian society: Land inequality and political rights. *The Journal of Conflict Resolution, 36*(3), 454–477.

Miller, M. (2023). Who values democracy? *Working paper*. https://doi.org/10.2139/ssrn.3488208.

Mizuno, N., Naito, K., & Okazawa, R. (2017). Inequality, extractive institutions, and growth in nondemocratic regimes. *Public Choice, 170*(1–2), 115–142.

Mounk, Y. (2018). *The People vs. Democracy: Why our freedom is in danger and how to save it*. Harvard University Press.

Muller, E. N. (1995). Economic determinants of democracy. *American Sociological Review, 60*(6), 966–982.

Nehru, V. (2015). Myanmar's military keeps firm grip on democratic transition. In *Carnegie endowment for international piece working paper*.

Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica, 49*(6), 1417–1426.

Piketty, T. (2014). *Capital in the twenty-first century*. Cambridge, MA: Belknap.

Policardo, L., & Carrera, E. J. S. (2020). Can income inequality promote democratization? *Metroeconomica, 71*(3), 510–532.

Ross, M. (2001). Does oil hinder democracy. *World Politics, 53*(3), 325–361.

Rowley, C. K., & Smith, N. (2009). Islam's democracy paradox: Muslims claim to like democracy, so why do they have so little? *Public Choice, 139*(3/4), 273–299.

Rød, E. G., Knutsen, C. H., & Hegre, H. (2020). The determinants of democracy: A sensitivity analysis. *Public Choice, 185*, 87–111.

Sabet, N. (2023). Turning out for redistribution: The effect of voter turnout on top marginal tax rates. *Public Choice, 194*, 347–367.

Skaaning, S.-E. (2018). Different types of data and the validity of democracy measures. *Politics and Governance, 6*(1), 105–116.

Treisman, D. (2020). Economic development and democracy: Predispositions and triggers. *Annual Review of Political Science, 23*(1), 241–257.

Tullock, G. (1986). Industrial organization and rent seeking in dictatorships. *Journal of Institutional and Theoretical Economics, 142*(1), 4–15.

Vaccaro, A. (2021). Comparing measures of democracy: Statistical properties, convergence, and interchangeability. *European Political Science, 20*, 666–684.

Wintrobe, R. (1990). The Tinpot and the totalitarian: An economic theory of dictatorship. *The American Political Science Review, 84*(3), 849–872.

Woodberry, R. (2012). The missionary roots of liberal democracy. *American Political Science Review, 106*(2), 244–274.

Ziblatt, D. (2008). Does landholding inequality block democratization?: A test of the "Bread and Democracy'' thesis and the case of Prussia. *World Politics, 60*(4), 610–641.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.